Contents

1 Ordinary Differential Equations (ODEs) & Linear Algebra

Welcome to the comprehensive study guide for **Ordinary Differential Equations (ODEs)**. This document serves as a digitized, structured, and easily navigable reference book based on handwritten lecture notes. It covers the fundamental concepts, methods of solving various differential equations, and their applications.

1.1 Course Learning Outcomes (CLOs)

Based on the course syllabus, by the end of this course, students will be able to:

- **CLO 1:** Describe the function of Differential Equations & systems of linear equations to explain physical situations. `[PLO 1]`
- **CLO 2:** Apply appropriate methods to solve differential equations & systems of linear equations relevant to engineering. `[PLO 2]`

1.2 Recommended Textbooks

To supplement the materials in this repository, the following textbooks are recommended:

1. **Elementary Linear Algebra** by *Howard Anton*
2. **A First Course in Differential Equations** by *Dennis G. Zill*
3. **Advanced Engineering Mathematics** by *Erwin Kreyszig*

1.3 Sessional Marks Criteria

Assessment Type	Quantity	Total Marks	Notes
Midterm Exam	1	20	
Assignments	2	15	
Quizzes	2	5	Best 2 are counted

1.4 Table of Contents

This document is organized into five modules for a complete learning path. Click on any section to jump directly to it.

1.4.1 Module 1: Fundamentals of Differential Equations

1.4.2 Module 2: Solving First-Order ODEs

Note: The broader course syllabus also includes Linear Algebra, Partial Differential Equations (PDEs), and Fourier Series. This document currently focuses on the digitized lecture notes for 1st and 2nd/Higher Order ODEs.

2 Module 1: Fundamentals of Differential Equations

Welcome to **Module 1**. Before diving into complex solving techniques, it is essential to understand the basic terminology, classifications, and origins of differential equations. This module lays the groundwork for the entire course.

2.1 Module Learning Objectives

By the end of this module, you will be able to:

1. Define a differential equation and distinguish between Ordinary (ODEs) and Partial (PDEs) Differential Equations.
2. Accurately identify the **Order** and **Degree** of any given differential equation.
3. Construct a differential equation from a general mathematical function by systematically eliminating arbitrary constants.

2.2 Topic Index

2.2.1 *1. Basic Definitions*

- **What is a Differential Equation?**
- **Ordinary Differential Equations (ODEs)** vs. **Partial Differential Equations (PDEs)**.
- Understanding dependent and independent variables.

2.2.2 *2. Order and Degree of a DE*

- **Order:** Identifying the highest differential coefficient.
- **Degree:** Determining the highest power of the highest derivative.
- **Rules:** Making the equation free from radicals and fractions before determining the degree.
- *Includes step-by-step solved exercises.*

2.2.3 *3. Formation of Differential Equations*

- The relationship between the number of arbitrary constants and the order of the resulting DE.
- **Methodology:** Differentiating ordinary equations and eliminating constants.
- *Includes step-by-step solved exercises.*

2.2.4 *4. Practice Questions*

- Problems covering definitions, order & degree, and formation of ODEs.
- *Includes fully worked solutions.*

3 Basic Definitions of Differential Equations

Before learning how to solve differential equations, it is crucial to understand what they are and how they are classified. This foundational knowledge will help you identify the type of equation you are dealing with, which dictates the method required to solve it.

3.1 1. What is a Differential Equation?

> **Definition:** A **Differential Equation (DE)** is a mathematical equation that relates one or more unknown functions and their derivatives.

In simpler terms, any equation that contains a "differential coefficient" (a derivative, such as $\frac{dy}{dx}$ or y') is called a differential equation. These equations are used extensively in physics, engineering, and biology to describe physical situations involving rates of change.

3.2 2. Classification of Differential Equations

Differential equations are broadly divided into two main categories based on the number of independent variables they contain:

3.2.1 2.1 Ordinary Differential Equations (ODEs)

> **Definition:** An ODE is an equation that contains one or several derivatives of an unknown function with respect to a **single independent variable**.

Typically, we deal with an unknown function $y(x)$, where:

- y is the **dependent variable**.
- x is the **independent variable**.

Examples of ODEs: Here are some standard examples of Ordinary Differential Equations:

1. A simple first-order ODE:
$$\frac{dy}{dx} = \cos x$$
2. A second-order ODE:
$$\frac{d^2y}{dx^2} + 9y = e^{-2x}$$
3. An ODE involving both variables in a rational expression:
$$\frac{dy}{dx} = \frac{1+x^2}{1-y^2}$$

3.2.2 2.2 Partial Differential Equations (PDEs)

> **Definition:** A PDE is a differential equation that contains unknown multivariable functions and their partial derivatives. This means there are **two or more independent variables**.

(Note: While PDEs are part of the broader course curriculum, the primary focus of the current study modules is on Ordinary Differential Equations).

Next Topic: Now that you know what a differential equation is, proceed to Order and Degree of a DE to learn how to classify ODEs by their highest derivatives and powers.

4 Order and Degree of a Differential Equation

Once you have identified an equation as a Differential Equation (DE), the next critical step is to classify it by its **Order** and **Degree**. These two properties determine the complexity of the equation and dictate which mathematical methods you will need to use to solve it.

4.1 1. Concept Overview

4.1.1 1.1 The Order of a Differential Equation

> **Definition:** The **Order** of a differential equation is the order of the highest differential coefficient (derivative) present in the equation.

- If the highest derivative is $\frac{dy}{dx}$ (or y'), it is a **1st-order** DE.

- If the highest derivative is $\frac{d^2y}{dx^2}$ (or y''), it is a **2nd-order** DE.
- If the highest derivative is $\frac{d^ny}{dx^n}$ (or $y^{(n)}$), it is an **n-th order** DE.

4.1.2 1.2 The Degree of a Differential Equation

> **Definition:** The **Degree** of a differential equation is the highest power (exponent) of the highest-order derivative present in the equation.

4.2 2. Important Working Rule

To accurately find the degree of a differential equation, the equation **must be made free from radicals and fractions** as far as the derivatives are concerned.

If a derivative is inside a square root or raised to a fractional power (e.g., 3/2), you must use algebraic manipulation (like squaring or cubing both sides) to clear those fractional powers before determining the degree.

4.3 3. Solved Questions

Here are step-by-step examples from the lecture notes demonstrating how to find the order and degree.

4.3.1 Question 1

Determine the order and degree of the following Differential Equation:

$$\cos x \frac{d^2y}{dx^2} + \sin x \left(\frac{dy}{dx}\right)^3 + 8y = \tan x$$

Solution:

1. **Find the Order:** Look for the highest derivative in the equation.
   - We have a first derivative: $\frac{dy}{dx}$
   - We have a second derivative: $\frac{d^2y}{dx^2}$
   - The highest derivative is the second derivative. Therefore, the **Order is 2**.
2. **Find the Degree:** Look at the power of the *highest* derivative.
   - The highest derivative is $\frac{d^2y}{dx^2}$.
   - It is not raised to any explicitly written power, which means its power is 1.
   - *(Note: Do not be tricked by the $\left(\frac{dy}{dx}\right)^3$ term. Even though it has a higher power of 3, it is not the highest-order derivative).*
3. **Final Answer:**
   - **Order:** 2
   - **Degree:** 1

4.3.2 Question 2

Determine the order and degree of the following Differential Equation:

$$\left[1 + \left(\frac{dy}{dx}\right)^2\right]^{3/2} = K\frac{d^2y}{dx^2}$$

Solution:

1. **Apply the Working Rule (Clear Radicals):** We notice that the derivatives on the left side are raised to a fractional power (3/2). We cannot determine the degree until this fraction is removed.
2. **Algebraic Manipulation:** To eliminate the denominator of the fraction (/2), we square both sides of the equation.

$$\left(\left[1 + \left(\frac{dy}{dx}\right)^2\right]^{3/2}\right)^2 = \left(K\frac{d^2y}{dx^2}\right)^2$$

$$\left[1 + \left(\frac{dy}{dx}\right)^2\right]^3 = K^2\left(\frac{d^2y}{dx^2}\right)^2$$

3. **Find the Order:** Now look at the manipulated equation. The highest derivative present is $\frac{d^2y}{dx^2}$.
   - Therefore, the **Order is 2**.
4. **Find the Degree:** Look at the power of that highest derivative.
   - The term $\left(\frac{d^2y}{dx^2}\right)$ is raised to the power of 2.
   - Therefore, the **Degree is 2**.
5. **Final Answer:**
   - **Order:** 2
   - **Degree:** 2

Next Topic: Now that you know how to classify DEs, proceed to Formation of Differential Equations to learn how these equations are originally constructed.

5 Formation of Differential Equations

Before learning the various techniques to solve Differential Equations (DEs), it is important to understand where they come from. A differential equation is typically formed by taking a standard mathematical equation (which represents a family of curves) and eliminating its arbitrary constants.

5.1 1. Concept Overview

Definition: A differential equation is formed by differentiating an ordinary equation and eliminating its arbitrary constants.

When you have an equation representing a family of curves, it will contain one or more unknown parameters, known as **arbitrary constants** (e.g., A, B, C, k). Your goal is to create a new equation that describes the *rate of change* of that family of curves without relying on those specific constants.

5.2 2. The Golden Rule of Formation

To know exactly how many times you need to differentiate the original equation, use this fundamental rule:

> **The Rule:** The number of arbitrary constants in the given general equation is exactly equal to the **Order** of the differential equation you need to form.
>
> - If the equation has **1** arbitrary constant, differentiate **only once**.
> - If the equation has **2** arbitrary constants, differentiate **twice**.
> - If the equation has n arbitrary constants, differentiate n **times**.

5.3 3. Solved Questions

Here are step-by-step examples from the lecture notes demonstrating how to form differential equations.

5.3.1 Question 1

Form the differential equation for the given equation by eliminating the arbitrary constant. Also, write its order and degree.

$$y = Ax + A^2$$

Solution:

1. **Identify the number of arbitrary constants:** There is only **1** arbitrary constant in this equation, which is A. Therefore, we must differentiate the equation only **once**.
2. **Differentiate with respect to x:**

$$\frac{dy}{dx} = A(1) + 0$$

   Let's denote $\frac{dy}{dx}$ as y'. So, $y' = A$.
3. **Eliminate the constant:** Now, substitute the value of A ($A = y'$) back into the original equation $y = Ax + A^2$.

$$y = x(y') + (y')^2$$

   Writing this in standard fractional notation:

$$y = x\frac{dy}{dx} + \left(\frac{dy}{dx}\right)^2$$

   (This is the required differential equation, free of the constant A.)
4. **Determine Order and Degree:**

- **Order:** The highest derivative is $\frac{dy}{dx}$, so the Order is **1**.
- **Degree:** The highest power of $\frac{dy}{dx}$ is 2, so the Degree is **2**.

5.3.2 Question 2

Form the differential equation for the given equation by eliminating the arbitrary constants.

$$y = A\cos x + B\sin x$$

Solution:

1. **Identify the number of arbitrary constants:** There are **2** arbitrary constants in this equation: A and B. Therefore, we must differentiate the equation **twice**.
2. **First Differentiation:** Differentiate y with respect to x:

$$y' = -A\sin x + B\cos x$$

3. **Second Differentiation:** Differentiate y' with respect to x to get the second derivative:

$$y'' = -A\cos x - B\sin x$$

4. **Eliminate the constants:** Notice that in the second derivative, we can factor out a negative sign:

$$y'' = -(A\cos x + B\sin x)$$

   Look closely at the expression inside the parentheses; it is exactly our original equation for y! Substitute y back in:

$$y'' = -y$$

   Rearranging to standard form gives us the final differential equation:

$$y'' + y = 0$$

5. **Determine Order and Degree (Bonus):**
   - **Order:** The highest derivative is y'' ($\frac{d^2y}{dx^2}$), so the Order is **2**.
   - **Degree:** The power of y'' is 1, so the Degree is **1**.

Next Steps: Test your understanding of these concepts with the Module 1 Practice Questions.

6 Module 1 Practice: Formation of Differential Equations

This section contains homework and practice problems to test your understanding of forming differential equations by eliminating arbitrary constants.

6.0.1 Question 1: Polynomial Function

Form the differential equation for: $y = ax + bx^2 + cx^3$

Solution:

1. **Identify constants:** There are 3 constants (a, b, c), so we must differentiate 3 times.

2. **Differentiate successively:**

   - $y' = a + 2bx + 3cx^2$
   - $y'' = 2b + 6cx$
   - $y''' = 6c \implies c = \frac{y'''}{6}$

3. **Back-substitute to eliminate constants:** Substitute c into the y'' equation to find b: $y'' = 2b + 6(\frac{y'''}{6})x \implies y'' = 2b + xy''' \implies b = \frac{1}{2}y'' - \frac{1}{2}xy'''$

   Substitute b and c into the y' equation to find a: $y' = a + 2(\frac{1}{2}y'' - \frac{1}{2}xy''')x + 3(\frac{y'''}{6})x^2$ $y' = a + xy'' - x^2y''' + \frac{1}{2}x^2y''' \implies a = y' - xy'' + \frac{1}{2}x^2y'''$

4. **Substitute a, b, c into the original equation:** $y = \left(y' - xy'' + \frac{1}{2}x^2y'''\right)x + \left(\frac{1}{2}y'' - \frac{1}{2}xy'''\right)x^2 + \left(\frac{y'''}{6}\right)x^3$ $y = xy' - x^2y'' + \frac{1}{2}x^3y''' + \frac{1}{2}x^2y'' - \frac{1}{2}x^3y''' + \frac{1}{6}x^3y'''$ $y = xy' - \frac{1}{2}x^2y'' + \frac{1}{6}x^3y'''$

5. **Simplify (Multiply by 6):** $6y = 6xy' - 3x^2y'' + x^3y'''$ **Final Answer:** $x^3y''' - 3x^2y'' + 6xy' - 6y = 0$

6.0.2 Question 2: Trigonometric Function

Form the differential equation for: $y = A\cos 2t + B\sin 2t$

Solution:

1. **Identify constants:** There are 2 constants (A, B), so we differentiate 2 times.
2. **First derivative (with respect to t):** $y' = -2A\sin 2t + 2B\cos 2t$
3. **Second derivative:** $y'' = -4A\cos 2t - 4B\sin 2t$
4. **Eliminate constants:** Notice that we can factor out a -4: $y'' = -4(A\cos 2t + B\sin 2t)$ The expression in the parentheses is exactly our original y. $y'' = -4y$ **Final Answer:** $y'' + 4y = 0$

6.0.3 Question 3: Exponential Function

Form the differential equation for: $y = ae^x + be^{2x} + ce^{3x}$

Solution: *Advanced Tip:* Instead of tedious substitution, we can use the reverse of the auxiliary equation method.

1. **Identify roots:** The terms e^{1x}, e^{2x}, e^{3x} indicate that the roots of the characteristic equation are $m_1 = 1$, $m_2 = 2$, and $m_3 = 3$.
2. **Form the characteristic equation:** $(m - 1)(m - 2)(m - 3) = 0$

3. **Expand the equation:** $(m^2-3m+2)(m-3) = 0$ $m^3-3m^2-3m^2+9m+2m-6 = 0$ $m^3 - 6m^2 + 11m - 6 = 0$
4. **Convert back to derivatives ($m^n \to y^{(n)}$): Final Answer:** $y'''-6y''+11y'-6y = 0$

6.0.4 *Question 4: Mixed Function*

Form the differential equation for: $y = ae^x + be^{-x} + c\cos x + d\sin x$

Solution:

1. **Identify roots:** We have 4 constants, so order is 4.
   - e^x and e^{-x} give roots $m = 1$ and $m = -1$.
   - $\cos 1x$ and $\sin 1x$ give complex conjugate roots $m = 0 \pm 1i \implies m = i$ and $m = -i$.
2. **Form the characteristic equation:** $(m-1)(m+1)(m-i)(m+i) = 0$
3. **Expand the equation (using difference of squares):** $(m^2-1)(m^2-i^2) = 0$ Since $i^2 = -1$: $(m^2-1)(m^2+1) = 0$ $m^4 - 1 = 0$
4. **Convert back to derivatives: Final Answer:** $y'''' - y = 0$ (or $\frac{d^4y}{dx^4} - y = 0$)

Next Module: You have successfully completed Module 1! You are now ready to begin solving equations. Proceed to Module 2: Solving First-Order ODEs.

7 Module 2: Solving First-Order Ordinary Differential Equations

Welcome to **Module 2**. In this module, we transition from analyzing the structure of differential equations to actually solving them. Specifically, we will focus on **First-Order, First-Degree Ordinary Differential Equations**.

There is no single "magic formula" to solve all differential equations. Instead, you must analyze the structure of the equation and select the appropriate mathematical method. This module covers the five standard methods for solving first-order DEs.

7.1 Module Learning Objectives

By the end of this module, you will be able to:

1. Identify the specific type or form of a given first-order differential equation.
2. Apply algebraic manipulation and substitutions to transform complex equations into solvable forms.
3. Master the five standard techniques for finding the general solution to a first-order DE.

7.2 Topic Index & Solution Methods

It is highly recommended to study these methods in this exact order, as later methods often reduce equations back down to earlier methods (e.g., Homogeneous equations reduce down to Variable Separable).

7.2.1 1. Variable Separable Method

- **The Concept:** Grouping all x terms with dx and all y terms with dy.
- **The Standard Form:** $f(y)dy = g(x)dx$
- *Includes 4 step-by-step solved questions.*

7.2.2 2. Homogeneous Differential Equations

- **The Concept:** Identifying functions where every term has the same total degree.
- **The Substitution:** Using $y = vx$ to reduce the equation to a Variable Separable form.
- *Includes 6 step-by-step solved questions.*

7.2.3 3. Equations Reducible to Homogeneous Form

- **The Concept:** Handling linear rational equations with constant terms: $\frac{dy}{dx} = \frac{ax+by+C}{Ax+By+C}$.
- **Case 1:** Ratios of coefficients are not equal ($\frac{a}{A} \neq \frac{b}{B}$).
- **Case 2:** Ratios of coefficients are equal ($\frac{a}{A} = \frac{b}{B}$).
- *Includes step-by-step solved questions.*

7.2.4 4. Linear Differential Equations

- **The Concept:** Solving equations of the form $\frac{dy}{dx} + P(x)y = Q(x)$.
- **The Tool:** Finding and applying the **Integrating Factor (I.F.)**.
- *Includes 2 step-by-step solved questions.*

7.2.5 5. Bernoulli's Equations (Reducible to Linear)

- **The Concept:** Solving non-linear equations of the form $\frac{dy}{dx} + P(x)y = Q(x)y^n$.
- **The Substitution:** Dividing by y^n and substituting $z = y^{1-n}$ to transform it into a standard Linear DE.
- *Includes 2 step-by-step solved questions.*

7.2.6 6. Practice Questions

- Categorized problems covering all five methods: Variable Separable, Homogeneous, Reducible to Homogeneous, Linear, and Bernoulli.
- *Includes fully worked solutions.*

8 1. Variable Separable Method

The simplest and most straightforward technique for solving first-order Ordinary Differential Equations is the **Variable Separable** method. Whenever you are faced with a new DE, you should always check if it can be solved using this method before attempting more complex techniques.

8.1 1. Concept Overview

Definition: A first-order differential equation is considered "separable" if it can be written entirely in the form:

$$f(y)dy = g(x)dx$$

This means you can completely isolate all terms containing the dependent variable y (including dy) on one side of the equals sign, and all terms containing the independent variable x (including dx) on the other side.

8.2 2. Working Rule (Step-by-Step)

To solve a differential equation using the variable separable method, follow these steps:

1. **Rearrange the Equation:** Use algebraic manipulation (factoring, cross-multiplying, dividing) to get all y-terms with dy on the left side, and all x-terms with dx on the right side.
2. **Integrate Both Sides:** Apply the integral sign to both sides of the separated equation: $\int f(y)dy = \int g(x)dx$.
3. **Add the Constant:** After integrating, add a single arbitrary constant of integration (usually C) to one side of the equation (conventionally the x-side).
4. **Apply Initial Conditions (If Given):** If the problem provides specific values for x and y (an Initial Value Problem), plug them into your general solution to solve for the exact value of C.

8.3 3. Solved Questions

Here are four questions from the lecture notes, demonstrating standard separation, factoring, initial value problems, and pre-substitution.

8.3.1 Question 1: Standard Separation

Solve: $(x+1)\frac{dy}{dx} = x(y^2+1)$

Solution:

1. **Separate the variables:** Divide by (y^2+1) and divide by $(x+1)$ to move terms:
$$\frac{1}{y^2+1}dy = \frac{x}{x+1}dx$$
2. **Manipulate the right side for easier integration:** Notice that $\frac{x}{x+1} = \frac{x+1-1}{x+1} = 1 - \frac{1}{x+1}$
$$\frac{1}{y^2+1}dy = \left(1 - \frac{1}{x+1}\right)dx$$
3. **Integrate both sides:**
$$\int \frac{1}{y^2+1}dy = \int \left(1 - \frac{1}{x+1}\right)dx$$
$$\tan^{-1}(y) = x - \ln|x+1| + C$$

4. **Final General Solution:**

$$y = \tan(x - \ln|x+1| + C)$$

8.3.2 Question 2: Factoring Before Separation

Solve: $(xy^2 + x)dx + (yx^2 + y)dy = 0$

Solution:

1. **Factor out common terms:**

$$x(y^2+1)dx + y(x^2+1)dy = 0$$

2. **Move the dy term to the right side:**

$$x(y^2+1)dx = -y(x^2+1)dy$$

3. **Separate variables:** Divide by (x^2+1) and (y^2+1):

$$\frac{x}{x^2+1}dx = -\frac{y}{y^2+1}dy$$

4. **Integrate both sides:** Multiply both sides by 2 to perfectly match the derivative of the denominators:

$$\frac{1}{2}\int \frac{2x}{x^2+1}dx = -\frac{1}{2}\int \frac{2y}{y^2+1}dy$$

$$\frac{1}{2}\ln|x^2+1| = -\frac{1}{2}\ln|y^2+1| + C$$

5. **Simplify using logarithm rules:**

$$\ln(x^2+1) + \ln(y^2+1) = 2C$$

$$\ln[(x^2+1)(y^2+1)] = C' \implies (x^2+1)(y^2+1) = e^{C'} = K$$

(Where K is just a new arbitrary constant).

8.3.3 Question 3: Initial Value Problem (IVP)

Solve: $x\frac{dy}{dx} + \cot y = 0$ given the initial condition $y = \frac{\pi}{4}$ when $x = \sqrt{2}$.

Solution:

1. **Separate the variables:**

$$x\frac{dy}{dx} = -\cot y$$

$$\frac{1}{\cot y}dy = -\frac{1}{x}dx \implies \tan y\, dy = -\frac{1}{x}dx$$

2. **Integrate both sides:**
$$\int \tan y \, dy = -\int \frac{1}{x} dx$$
$$-\ln|\cos y| = -\ln|x| + \ln C$$
(Using $\ln C$ *instead of* C *makes simplification easier).*
3. **Simplify the General Solution:** Multiply by -1:
$$\ln|\cos y| = \ln|x| - \ln C \implies \ln(\cos y) = \ln\left(\frac{x}{C}\right)$$
$$\cos y = \frac{x}{C} \implies x = C\cos y$$
4. **Apply Initial Conditions to find C:** Substitute $x = \sqrt{2}$ and $y = \pi/4$:
$$\sqrt{2} = C\cos\left(\frac{\pi}{4}\right)$$
$$\sqrt{2} = C\left(\frac{1}{\sqrt{2}}\right) \implies C = \sqrt{2} \times \sqrt{2} = 2$$
5. **Final Particular Solution:**
$$x = 2\cos y \quad \text{or} \quad y = \cos^{-1}\left(\frac{x}{2}\right)$$

8.3.4 Question 4: Substitution Reducing to Separable

Solve: $x^4\frac{dy}{dx} + x^3y = -\sec(xy)$

Solution:

1. **Factor out x^3 on the left side:**
$$x^3\left(x\frac{dy}{dx} + y\right) = -\sec(xy)$$
2. **Notice the product rule derivative:** The term $(x\frac{dy}{dx}+y)$ is exactly the derivative of the product (xy). Let's use a substitution.
   - Let $t = xy$
   - Differentiate with respect to x: $\frac{dt}{dx} = x\frac{dy}{dx} + y(1) = x\frac{dy}{dx} + y$
3. **Substitute t and dt/dx into the equation:**
$$x^3\frac{dt}{dx} = -\sec t$$
4. **Now it is Variable Separable!**
$$\frac{1}{\sec t}dt = -\frac{1}{x^3}dx \implies \cos t \, dt = -x^{-3}dx$$

5. **Integrate:**

$$\int \cos t \, dt = -\int x^{-3} dx$$

$$\sin t = -\left(\frac{x^{-2}}{-2}\right) + C$$

$$\sin t = \frac{1}{2x^2} + C$$

6. **Resubstitute $t = xy$ for the final solution:**

$$\sin(xy) = \frac{1}{2x^2} + C$$

Next Method: What happens if the variables *cannot* be cleanly separated? Proceed to Method 2: Homogeneous Differential Equations to learn the next technique.

9 2. Homogeneous Differential Equations

When the variables in a first-order differential equation cannot be separated directly (i.e., you cannot get all x's on one side and y's on the other), the next method to check is whether the equation is **Homogeneous**.

9.1 1. Concept Overview

9.1.1 1.1 What is a Homogeneous Function?

A function $F(x, y)$ is called homogeneous of degree n if, when you replace x with λx and y with λy, the parameter λ can be factored out completely as λ^n.

$$F(\lambda x, \lambda y) = \lambda^n F(x, y)$$

9.1.2 1.2 What is a Homogeneous Differential Equation?

A first-order differential equation $\frac{dy}{dx} = f(x, y)$ is homogeneous if the function $f(x, y)$ can be expressed as a ratio of two homogeneous functions of the **same degree**.

Alternatively, if you can write the differential equation in the form:

$$\frac{dy}{dx} = F\left(\frac{y}{x}\right)$$

then it is a Homogeneous Differential Equation.

9.2 2. Working Rule (The Substitution Method)

To solve a homogeneous differential equation, we use a specific substitution to transform it into a **Variable Separable** equation.

1. **Identify Homogeneity:** Check that every term in the numerator and denominator has the same total power of variables.
2. **Substitute:**

- Let $y = vx$
- Differentiate with respect to x (using the product rule):

$$\frac{dy}{dx} = v + x\frac{dv}{dx}$$

3. **Transform:** Replace y and $\frac{dy}{dx}$ in the original equation. The x terms will typically cancel out, leaving an equation in terms of v and x.
4. **Separate and Integrate:** The new equation will be Variable Separable. Separate v and x, then integrate both sides.
5. **Resubstitute:** Finally, replace v with $\frac{y}{x}$ to get the general solution in terms of x and y.

9.3 3. Solved Questions

Here are questions from the lecture notes demonstrating the homogeneity check and the solution process.

9.3.1 Question 1: Checking for Homogeneity

Determine if the function $F(x, y) = \frac{2x^2y}{x^2+y^2}$ is homogeneous.

Solution:

1. **Test:** Replace $x \to \lambda x$ and $y \to \lambda y$.

$$F(\lambda x, \lambda y) = \frac{2(\lambda x)^2(\lambda y)}{(\lambda x)^2 + (\lambda y)^2}$$

2. **Simplify:**

$$= \frac{2\lambda^2x^2 \cdot \lambda y}{\lambda^2x^2 + \lambda^2y^2}$$
$$= \frac{\lambda^3(2x^2y)}{\lambda^2(x^2 + y^2)}$$

3. **Factor:**

$$= \lambda^{3-2}\left(\frac{2x^2y}{x^2 + y^2}\right) = \lambda^1 F(x, y)$$

4. **Conclusion:** Yes, it is a homogeneous function of degree **1**.

9.3.2 Question 2: Solving a Homogeneous DE

Solve: $(2xy + x^2)\frac{dy}{dx} = 3y^2 + 2xy$

Solution:

1. **Rewrite in standard form:**

$$\frac{dy}{dx} = \frac{3y^2 + 2xy}{2xy + x^2}$$

Observation: Every term (y^2, xy, x^2) is of degree 2. Thus, it is Homogeneous.

2. **Apply Substitution:** Let $y = vx$, so $\frac{dy}{dx} = v + x\frac{dv}{dx}$. Substitute into the equation:

$$v + x\frac{dv}{dx} = \frac{3(vx)^2 + 2x(vx)}{2x(vx) + x^2}$$

3. **Simplify the Fraction:**

$$v + x\frac{dv}{dx} = \frac{x^2(3v^2 + 2v)}{x^2(2v + 1)}$$

$$v + x\frac{dv}{dx} = \frac{3v^2 + 2v}{2v + 1}$$

4. **Separate Variables:** Move v to the right side:

$$x\frac{dv}{dx} = \frac{3v^2 + 2v}{2v + 1} - v$$

Find a common denominator:

$$x\frac{dv}{dx} = \frac{3v^2 + 2v - v(2v + 1)}{2v + 1}$$

$$x\frac{dv}{dx} = \frac{3v^2 + 2v - 2v^2 - v}{2v + 1} = \frac{v^2 + v}{2v + 1}$$

Now, separate v and x:

$$\frac{2v + 1}{v^2 + v}dv = \frac{1}{x}dx$$

5. **Integrate:**

$$\int \frac{2v + 1}{v^2 + v}dv = \int \frac{1}{x}dx$$

Notice: The numerator $(2v + 1)$ is the exact derivative of the denominator $(v^2 + v)$.

$$\ln|v^2 + v| = \ln|x| + \ln C$$

$$v^2 + v = Cx$$

6. **Resubstitute $v = y/x$:**

$$\left(\frac{y}{x}\right)^2 + \frac{y}{x} = Cx$$

$$\frac{y^2 + xy}{x^2} = Cx \implies y^2 + xy = Cx^3$$

9.3.3 Question 3: Trigonometric Homogeneous DE

Solve: $x\sin\left(\frac{y}{x}\right)dy = \left[y\sin\left(\frac{y}{x}\right) - x\right]dx$

Solution:

1. **Rewrite in standard form:**
$$\frac{dy}{dx} = \frac{y\sin(y/x) - x}{x\sin(y/x)}$$
$$\frac{dy}{dx} = \frac{y}{x} - \frac{x}{x\sin(y/x)} = \frac{y}{x} - \frac{1}{\sin(y/x)}$$
Observation: The equation is a function of (y/x), so it is Homogeneous.
2. **Apply Substitution:** Let $y = vx$, so $\frac{dy}{dx} = v + x\frac{dv}{dx}$.
$$v + x\frac{dv}{dx} = v - \frac{1}{\sin v}$$
3. **Simplify:** Subtract v from both sides:
$$x\frac{dv}{dx} = -\frac{1}{\sin v} = -\csc v$$
4. **Separate Variables:**
$$\sin v \, dv = -\frac{1}{x}dx$$
5. **Integrate:**
$$\int \sin v \, dv = -\int \frac{1}{x}dx$$
$$-\cos v = -\ln|x| + C$$
Multiply by -1:
$$\cos v = \ln|x| - C$$
6. **Resubstitute $v = y/x$:**
$$\cos\left(\frac{y}{x}\right) = \ln|x| + C'$$

Next Method: What if the equation looks homogeneous but has constant terms interfering? Proceed to Method 3: Equations Reducible to Homogeneous Form.

10 3. Equations Reducible to Homogeneous Form

Sometimes, a differential equation looks almost homogeneous but contains constant terms that break the symmetry. These equations usually describe the intersection of non-homogeneous lines. We can “reduce” them to a homogeneous form (or directly to variable separable form) using specific substitutions.

10.1 1. Concept Overview

This method applies to differential equations of the form:
$$\frac{dy}{dx} = \frac{ax + by + c}{Ax + By + C}$$

Here, the constants c and C prevent the equation from being homogeneous. To solve this, we compare the ratios of the coefficients of x and y.

There are two distinct cases:

10.1.1 Case 1: Intersecting Lines ($\frac{a}{A} \neq \frac{b}{B}$)

If the ratios are **not equal**, the lines represented by the numerator and denominator intersect at a specific point (h, k).

- **Substitution:** We shift the origin to that intersection point.
  - $x = X + h$
  - $y = Y + k$
  - $dy = dY, dx = dX$
- We find h and k by solving the system of equations $ax+by+c = 0$ and $Ax+By+C = 0$.
- This removes the constants c and C, resulting in a standard Homogeneous DE in variables X and Y.

10.1.2 Case 2: Parallel Lines ($\frac{a}{A} = \frac{b}{B}$)

If the ratios **are equal**, the lines are parallel. This means the terms $(ax + by)$ and $(Ax + By)$ are scalar multiples of each other.

- **Substitution:** We substitute the common linear expression with a new variable z.
  - Let $z = ax + by$
  - Differentiate with respect to x to find $\frac{dz}{dx}$.
- This immediately reduces the DE to a **Variable Separable** form.

10.2 2. Solved Questions

Here is a step-by-step solution for a **Case 2** problem found in the lecture notes.

10.2.1 Question 1 (Case 2: Equal Ratios)

Solve: $(x + 2y)(dx - dy) = dx + dy$

Solution:

Step 1: Rewrite in Standard Form First, rearrange the equation to isolate $\frac{dy}{dx}$. Expand the terms:

$$x\,dx - x\,dy + 2y\,dx - 2y\,dy = dx + dy$$

Group dx and dy terms:

$$(x + 2y)\,dx - dx = (x + 2y)\,dy + dy$$

$$(x + 2y - 1)\,dx = (x + 2y + 1)\,dy$$

$$\frac{dy}{dx} = \frac{x + 2y - 1}{x + 2y + 1}$$

Step 2: Check Ratios Compare coefficients of x and y in the numerator $(a = 1, b = 2)$ and denominator $(A = 1, B = 2)$.

$$\frac{a}{A} = \frac{1}{1} = 1, \quad \frac{b}{B} = \frac{2}{2} = 1$$

Since $\frac{a}{A} = \frac{b}{B}$, this is **Case 2**.

Step 3: Apply Substitution Let the common term be z.

$$z = x + 2y$$

Differentiate with respect to x:

$$\frac{dz}{dx} = 1 + 2\frac{dy}{dx} \implies \frac{dy}{dx} = \frac{1}{2}\left(\frac{dz}{dx} - 1\right)$$

Step 4: Substitute into the DE Replace $\frac{dy}{dx}$ and $(x+2y)$ in the standard form equation:

$$\frac{1}{2}\left(\frac{dz}{dx} - 1\right) = \frac{z-1}{z+1}$$

Multiply by 2:

$$\frac{dz}{dx} - 1 = \frac{2(z-1)}{z+1} = \frac{2z-2}{z+1}$$

Add 1 to both sides:

$$\frac{dz}{dx} = \frac{2z-2}{z+1} + 1$$

Find common denominator:

$$\frac{dz}{dx} = \frac{2z-2+(z+1)}{z+1} = \frac{3z-1}{z+1}$$

Step 5: Separate Variables and Integrate

$$\frac{z+1}{3z-1}dz = dx$$

Integrate both sides:

$$\int \frac{z+1}{3z-1}dz = \int dx$$

Integration Tip: To integrate the left side, make the numerator resemble the derivative of the denominator or perform long division. Multiply and divide by 3 inside the integral for easier manipulation:

$$\frac{1}{3}\int \frac{3z+3}{3z-1}dz = x + C$$

$$\frac{1}{3}\int \frac{(3z-1)+4}{3z-1}dz = x + C$$

$$\frac{1}{3}\int \left(1 + \frac{4}{3z-1}\right)dz = x + C$$

$$\frac{1}{3}\left[z + 4 \cdot \frac{\ln|3z-1|}{3}\right] = x + C$$

Step 6: Final Substitution Multiply by 3 to clear the fraction:

$$z + \frac{4}{3}\ln|3z-1| = 3x + 3C$$

Substitute $z = x + 2y$ back:

$$(x + 2y) + \frac{4}{3} \ln |3(x + 2y) - 1| = 3x + C'$$

Rearrange for the final implicit solution:

$$2y - 2x + \frac{4}{3} \ln |3x + 6y - 1| = C'$$

Next Method: While Homogeneous and Reducible methods deal with ratios, many physical systems are modeled by **Linear Differential Equations**. Proceed to Method 4: Linear ODEs.

11 4. Linear Differential Equations

Linear Differential Equations are among the most important types of ODEs because they appear frequently in physics and engineering (e.g., circuits, mixing problems, cooling laws). Unlike previous methods where we manipulated variables, here we follow a strict algorithmic formula.

11.1 1. Concept Overview

A first-order differential equation is **Linear** if:

1. The dependent variable (y) and its derivative ($\frac{dy}{dx}$) occur only in the **first degree** (power is 1).
2. There are **no products** of the dependent variable and its derivative (e.g., no $y \cdot \frac{dy}{dx}$ terms).
3. There are no transcendental functions of y (like $\sin y$, e^y, $\ln y$).

11.1.1 The Standard Form

A linear differential equation in y is written as:

$$\frac{dy}{dx} + P(x)y = Q(x)$$

Where $P(x)$ and $Q(x)$ are functions of x alone (or constants).

11.2 2. Working Rule (Integrating Factor Method)

To solve a linear equation, we use an **Integrating Factor (I.F.)**, which transforms the left side of the equation into a perfect product derivative.

Step 1: Standardize Ensure the coefficient of $\frac{dy}{dx}$ is **1**. If there is a term multiplied by $\frac{dy}{dx}$, divide the entire equation by that term.

Step 2: Identify P and Q Compare your equation to the standard form and identify $P(x)$ and $Q(x)$.

Step 3: Find the Integrating Factor (I.F.) Calculate the exponential integral of $P(x)$:

$$I.F. = e^{\int P(x)dx}$$

Step 4: Write the General Solution The solution is given by the formula:

$$y \cdot (I.F.) = \int [Q(x) \cdot (I.F.)]dx + C$$

Step 5: Evaluate the Integral Solve the integral on the right-hand side to find the final answer.

11.3 3. Solved Questions

Here are two step-by-step examples from the lecture notes.

11.3.1 Question 1: Algebraic P(x)

Solve: $(x-1)^3\frac{dy}{dx} + 4(x-1)^2y = x+1$

Solution:

Step 1: Standardize The coefficient of $\frac{dy}{dx}$ is $(x-1)^3$. We must divide the entire equation by $(x-1)^3$ to isolate $\frac{dy}{dx}$.

$$\frac{dy}{dx} + \frac{4(x-1)^2}{(x-1)^3}y = \frac{x+1}{(x-1)^3}$$

$$\frac{dy}{dx} + \frac{4}{x-1}y = \frac{x+1}{(x-1)^3}$$

Step 2: Identify P and Q Comparing to $\frac{dy}{dx} + Py = Q$:

- $P(x) = \frac{4}{x-1}$
- $Q(x) = \frac{x+1}{(x-1)^3}$

Step 3: Calculate I.F.

$$I.F. = e^{\int \frac{4}{x-1}dx} = e^{4\ln(x-1)}$$

Using logarithm laws ($a \ln b = \ln b^a$):

$$I.F. = e^{\ln((x-1)^4)} = (x-1)^4$$

Step 4: Apply General Solution Formula

$$y \cdot (x-1)^4 = \int \left[\frac{x+1}{(x-1)^3} \cdot (x-1)^4\right] dx + C$$

Step 5: Integrate Simplify the term inside the integral:

$$y(x-1)^4 = \int (x+1)(x-1)dx + C$$

Using difference of squares $(a+b)(a-b) = a^2 - b^2$:

$$y(x-1)^4 = \int (x^2-1)dx + C$$

$$y(x-1)^4 = \frac{x^3}{3} - x + C$$

11.3.2 Question 2: Trigonometric P(x)

Solve: $\frac{dy}{dx} + y\tan x = \cos^2 x$

Solution:

Step 1: Standardize The coefficient of $\frac{dy}{dx}$ is already 1. No division needed.

Step 2: Identify P and Q

- $P(x) = \tan x$
- $Q(x) = \cos^2 x$

Step 3: Calculate I.F.

$$I.F. = e^{\int \tan x dx}$$

Recall standard integral: $\int \tan x dx = \ln|\sec x|$.

$$I.F. = e^{\ln(\sec x)} = \sec x$$

Step 4: Apply General Solution Formula

$$y \cdot (\sec x) = \int [\cos^2 x \cdot \sec x] dx + C$$

Step 5: Integrate Simplify the term inside the integral (since $\sec x = 1/\cos x$):

$$y \sec x = \int \left(\cos^2 x \cdot \frac{1}{\cos x}\right) dx + C$$

$$y \sec x = \int \cos x dx + C$$

$$y \sec x = \sin x + C$$

Final Answer: You can leave it as is, or multiply by $\cos x$:

$$y = \sin x \cos x + C \cos x$$

Next Method: Sometimes an equation is *almost* linear but has a pesky y^n term on the right side. This is called a **Bernoulli Equation**. Proceed to Method 5: Bernoulli's Equations.

12 5. Bernoulli's Equations (Reducible to Linear)

Bernoulli's equations are a specific class of **non-linear** differential equations that can be easily transformed into a standard **Linear** differential equation using a clever substitution. This method is the final standard technique for first-order ODEs.

12.1 1. Concept Overview

A Bernoulli differential equation has the form:

$$\frac{dy}{dx} + P(x)y = Q(x)y^n$$

Where n is any real number except 0 or 1.

- If $n = 0$, the equation is Linear.
- If $n = 1$, the equation is Variable Separable.
- For any other n, the term y^n on the right side makes the equation **non-linear**.

12.2 2. Working Rule (Reduction to Linear Form)

To solve a Bernoulli equation, we must "linearize" it.

Step 1: Divide by the non-linear term Divide the entire equation by y^n to move the y terms to the left side:

$$y^{-n}\frac{dy}{dx} + P(x)y^{1-n} = Q(x)$$

Step 2: Substitute Let $z = y^{1-n}$ (the term attached to $P(x)$).

Step 3: Differentiate Differentiate z with respect to x:

$$\frac{dz}{dx} = (1-n)y^{-n}\frac{dy}{dx}$$

Rearrange this to match the first term of your divided equation:

$$\frac{1}{1-n}\frac{dz}{dx} = y^{-n}\frac{dy}{dx}$$

Step 4: Transform into Linear DE Substitute z and $\frac{dz}{dx}$ back into the equation. It will now be a standard Linear DE in terms of z and x:

$$\frac{dz}{dx} + (1-n)P(x)z = (1-n)Q(x)$$

Step 5: Solve and Resubstitute Solve this new linear equation for z using the Integrating Factor method (see Linear Differential Equations). Finally, replace z with y^{1-n} to get the general solution.

12.3 3. Solved Questions

Here are two step-by-step examples from the lecture notes.

12.3.1 Question 1: Rational Powers

Solve: $\frac{dy}{dx} + \frac{y}{x} = \frac{y^2}{x^2}$

Solution:

Step 1: Identify Form This matches Bernoulli's form with $P(x) = 1/x$, $Q(x) = 1/x^2$, and $n = 2$.

Step 2: Divide by y^n (y^2)

$$y^{-2}\frac{dy}{dx} + \frac{1}{x}y^{-1} = \frac{1}{x^2}$$

Step 3: Substitute Let $z = y^{1-2} = y^{-1}$. Differentiate: $\frac{dz}{dx} = -1 \cdot y^{-2}\frac{dy}{dx} \implies -\frac{dz}{dx} = y^{-2}\frac{dy}{dx}$.

Step 4: Transform Substitute into the equation:

$$-\frac{dz}{dx} + \frac{1}{x}z = \frac{1}{x^2}$$

Multiply by -1 to standardize:

$$\frac{dz}{dx} - \frac{1}{x}z = -\frac{1}{x^2}$$

Step 5: Solve Linear DE

- **I.F.:** $e^{\int -\frac{1}{x}dx} = e^{-\ln x} = \frac{1}{x}$
- **Solution Formula:**

$$z \cdot \left(\frac{1}{x}\right) = \int \left[-\frac{1}{x^2} \cdot \frac{1}{x}\right] dx + C$$

$$\frac{z}{x} = -\int x^{-3}dx + C$$

$$\frac{z}{x} = -\left(\frac{x^{-2}}{-2}\right) + C = \frac{1}{2x^2} + C$$

Step 6: Resubstitute $(z = 1/y)$

$$\frac{1}{xy} = \frac{1}{2x^2} + C$$

12.3.2 Question 2: Cubic Power

Solve: $\frac{dy}{dx} + 4xy = xy^3$

Solution:

Step 1: Identify Form This is Bernoulli's form with $n = 3$.

Step 2: Divide by y^3

$$y^{-3}\frac{dy}{dx} + 4xy^{-2} = x$$

Step 3: Substitute Let $z = y^{1-3} = y^{-2}$. Differentiate: $\frac{dz}{dx} = -2y^{-3}\frac{dy}{dx} \implies -\frac{1}{2}\frac{dz}{dx} = y^{-3}\frac{dy}{dx}$.

Step 4: Transform Substitute into the equation:

$$-\frac{1}{2}\frac{dz}{dx} + 4xz = x$$

Multiply by -2 to standardize:

$$\frac{dz}{dx} - 8xz = -2x$$

Step 5: Solve Linear DE

- **Identify P(x):** $P(x) = -8x$
- **I.F.:** $e^{\int -8xdx} = e^{-4x^2}$
- **Solution Formula:**
$$z \cdot e^{-4x^2} = \int (-2x)e^{-4x^2} dx + C$$

Step 6: Integrate To solve $\int -2xe^{-4x^2} dx$, use substitution $u = -4x^2 \implies du = -8xdx \implies \frac{1}{4}du = -2xdx$.

$$\text{Right Side} = \int e^u(\frac{1}{4}du) = \frac{1}{4}e^u = \frac{1}{4}e^{-4x^2}$$

So the equation becomes:

$$ze^{-4x^2} = \frac{1}{4}e^{-4x^2} + C$$

Step 7: Resubstitute $(z = y^{-2})$

$$y^{-2}e^{-4x^2} = \frac{1}{4}e^{-4x^2} + C$$

Multiply by e^{4x^2}:

$$\frac{1}{y^2} = \frac{1}{4} + Ce^{4x^2}$$

Next Steps: Test your mastery of all five first-order methods with the Module 2 Practice Questions.

13 Module 2 Practice: Solving First-Order ODEs

This section contains practice questions from your homework assignments, categorized by the specific solving technique required.

13.1 Part A: Variable Separable Method

13.1.1 Question 1

Solve: $(1 + x^2)dy - xy\, dx = 0$

Solution:

1. **Separate Variables:** Move the dx term to the right: $(1 + x^2)dy = xy\, dx$ Divide by y and $(1 + x^2)$: $\frac{1}{y}dy = \frac{x}{1+x^2}dx$

2. **Integrate both sides:** $\int \frac{1}{y}dy = \frac{1}{2}\int \frac{2x}{1+x^2}dx$ *(multiplied by 2 to match derivative of denominator)* $\ln|y| = \frac{1}{2}\ln(1+x^2) + \ln C$
3. **Simplify:** $\ln|y| = \ln\left(C\sqrt{1+x^2}\right)$ $y = C\sqrt{1+x^2}$ Squaring both sides gives the **Final Answer:** $y^2 = C^2(1+x^2)$

13.1.2 Question 2

Solve: $(e^y + 2)\sin x\,dx - e^y\cos x\,dy = 0$

Solution:

1. **Separate Variables:** $(e^y + 2)\sin x\,dx = e^y\cos x\,dy$ $\frac{\sin x}{\cos x}dx = \frac{e^y}{e^y+2}dy$
2. **Integrate both sides:** $\int \tan x\,dx = \int \frac{e^y}{e^y+2}dy$ $-\ln|\cos x| = \ln|e^y + 2| - \ln C$
3. **Simplify:** $\ln C = \ln|e^y + 2| + \ln|\cos x| \implies \ln C = \ln((e^y + 2)\cos x)$ **Final Answer:** $(e^y + 2)\cos x = C$

13.2 Part B: Homogeneous Equations

13.2.1 Question 3

Solve: $(y^2 - xy)dx + x^2dy = 0$

Solution:

1. **Standard Form:** $x^2dy = -(y^2 - xy)dx \implies \frac{dy}{dx} = \frac{xy-y^2}{x^2}$
2. **Substitute:** Let $y = vx \implies \frac{dy}{dx} = v + x\frac{dv}{dx}$ $v + x\frac{dv}{dx} = \frac{x(vx)-(vx)^2}{x^2} = \frac{x^2(v-v^2)}{x^2} = v - v^2$
3. **Separate and Integrate:** $x\frac{dv}{dx} = v - v^2 - v \implies x\frac{dv}{dx} = -v^2$ $-\frac{1}{v^2}dv = \frac{1}{x}dx$ $\int -v^{-2}dv = \int \frac{1}{x}dx \implies \frac{1}{v} = \ln|x| + C$
4. **Resubstitute** $v = y/x$**:** $\frac{x}{y} = \ln|x| + C$ **Final Answer:** $x = y(\ln|x| + C)$

13.3 Part C: Linear Differential Equations

13.3.1 Question 4

Solve: $\frac{dy}{dx} + \frac{1}{x}y = x^3 - 3$

Solution:

1. **Identify P and Q:** $P(x) = \frac{1}{x}$, $Q(x) = x^3 - 3$
2. **Find Integrating Factor (I.F):** $I.F. = e^{\int \frac{1}{x}dx} = e^{\ln x} = x$
3. **General Solution Formula:** $y \cdot (x) = \int x(x^3 - 3)dx + C$ $xy = \int(x^4 - 3x)dx + C$ **Final Answer:** $xy = \frac{x^5}{5} - \frac{3x^2}{2} + C$

13.4 Part D: Bernoulli's Equations

13.4.1 Question 5

Solve: $2x\frac{dy}{dx} = 10x^3y^5 + y$

Solution:

1. **Standard Form:** $2x\frac{dy}{dx} - y = 10x^3y^5 \implies \frac{dy}{dx} - \frac{1}{2x}y = 5x^2y^5$

2. **Divide by y^5:** $y^{-5}\frac{dy}{dx} - \frac{1}{2x}y^{-4} = 5x^2$
3. **Substitute:** Let $z = y^{-4} \implies \frac{dz}{dx} = -4y^{-5}\frac{dy}{dx} \implies -\frac{1}{4}\frac{dz}{dx} = y^{-5}\frac{dy}{dx}$ $-\frac{1}{4}\frac{dz}{dx} - \frac{1}{2x}z = 5x^2$ Multiply by -4: $\frac{dz}{dx} + \frac{2}{x}z = -20x^2$ (This is Linear)
4. **Solve Linear DE for z:** $I.F. = e^{\int \frac{2}{x}dx} = e^{2\ln x} = x^2$ $z \cdot x^2 = \int(-20x^2 \cdot x^2)dx + C = \int -20x^4 dx + C$ $zx^2 = -4x^5 + C$
5. **Resubstitute $z = y^{-4}$:** $y^{-4}x^2 = -4x^5 + C$ **Final Answer:** $\frac{x^2}{y^4} = -4x^5 + C$

Next Module: Now that we have mastered all methods for solving first-order equations, we will explore their geometric applications. Proceed to Module 3: Applications of First-Order ODEs.

14 Module 3: Applications of First-Order ODEs

Welcome to **Module 3**. Now that you have mastered the techniques for solving first-order differential equations, we can apply these skills to solve geometric and physical problems.

Differential equations are not just abstract mathematical constructs; they are the language used to describe the physical universe, from the motion of planets to the flow of electricity. In this module, we focus on a specific geometric application found in your course notes: **Orthogonal Trajectories**.

14.1 Module Learning Objectives

By the end of this module, you will be able to:

1. Understand the geometric relationship between families of curves.
2. Calculate the **Orthogonal Trajectories** of a given family of curves (curves that intersect at right angles).

14.2 Topic Index

14.2.1 1. Orthogonal Trajectories

- **The Concept:** What does it mean for two families of curves to be orthogonal?
- **The Method:** Using the negative reciprocal of the slope ($-\frac{dx}{dy}$) to find the perpendicular family.
- *Includes step-by-step solved questions for algebraic and trigonometric curves.*

14.2.2 2. Practice Questions

- Problems on finding orthogonal trajectories for circles, parabolas, and exponential curves.
- *Includes fully worked solutions.*

15 Orthogonal Trajectories

In geometry and physics, we often deal with families of curves that interact with each other. One of the most common interactions is when two families of curves intersect each other at right angles (90°). These are called **Orthogonal Trajectories**.

Common examples include:

- **Electric Fields:** Equipotential lines and electric field lines are orthogonal trajectories of each other.
- **Heat Flow:** Isotherms (lines of constant temperature) and lines of heat flow are orthogonal.

15.1 1. Concept Overview

Definition: An **Orthogonal Trajectory** is a curve that intersects every curve of a given family at right angles.

If we have a family of curves represented by $F(x, y, c) = 0$, its "slope" at any point is given by the derivative $\frac{dy}{dx}$ (let's call this slope m_1).

For a second curve to be perpendicular (orthogonal) to the first, its slope (m_2) must be the **negative reciprocal** of m_1:

$$m_1 \cdot m_2 = -1 \implies m_2 = -\frac{1}{m_1}$$

Therefore, the differential equation for the orthogonal trajectory is found by replacing $\frac{dy}{dx}$ with $-\frac{dx}{dy}$.

15.2 2. Working Rule (Step-by-Step)

To find the orthogonal trajectories of a given family of curves:

1. **Find the Differential Equation:** Differentiate the given equation of the family of curves with respect to x to find $\frac{dy}{dx}$.
2. **Eliminate the Parameter:** If the derivative still contains the arbitrary constant (parameter C or k), use the original equation to substitute it out. The differential equation must only contain $x, y, \frac{dy}{dx}$.
3. **Apply Orthogonality Condition:** Replace $\frac{dy}{dx}$ with $-\frac{dx}{dy}$ in the differential equation.
$$\frac{dy}{dx} \longrightarrow -\frac{dx}{dy}$$
4. **Solve the New Equation:** Solve this new differential equation (usually by Variable Separation) to find the family of orthogonal trajectories.

15.3 3. Solved Questions

Here are two examples from the lecture notes illustrating this process.

15.3.1 Question 1: Hyperbolas

Find the orthogonal trajectories of the family of curves:

$$x^2 - y^2 = C$$

(This represents a family of rectangular hyperbolas).

Solution:

Step 1: Differentiate the Given Equation Differentiate with respect to x:

$$2x - 2y\frac{dy}{dx} = 0$$

$$2y\frac{dy}{dx} = 2x$$

$$\frac{dy}{dx} = \frac{x}{y}$$

(Note: The constant C disappeared automatically upon differentiation).

Step 2: Apply Orthogonality Condition To find the orthogonal path, replace $\frac{dy}{dx}$ with $-\frac{dx}{dy}$:

$$-\frac{dx}{dy} = \frac{x}{y}$$

Step 3: Solve the New Differential Equation Separate the variables:

$$-\frac{dx}{x} = \frac{dy}{y}$$

Integrate both sides:

$$-\int \frac{1}{x}dx = \int \frac{1}{y}dy$$

$$-\ln|x| = \ln|y| + \ln K$$

(Using $\ln K$ as the integration constant for easier simplification).

Step 4: Simplify

$$\ln|y| + \ln|x| = -\ln K$$

$$\ln|xy| = C'$$

$$xy = \text{Constant}$$

Result: The orthogonal trajectories of the hyperbolas $x^2 - y^2 = C$ are the hyperbolas $xy = K$.

15.3.2 Question 2: Parabolas

Find the orthogonal trajectories of the family of curves:

$$y = (x - k)^2$$

(This represents a family of parabolas shifted along the x-axis).

Solution:

Step 1: Differentiate the Given Equation Differentiate with respect to x:

$$\frac{dy}{dx} = 2(x - k)$$

Step 2: Eliminate the Parameter (k) Our derivative depends on k, which is not allowed in the final DE. From the original equation $y = (x-k)^2$, we can say:

$$x - k = \sqrt{y}$$

Substitute this back into the derivative:

$$\frac{dy}{dx} = 2\sqrt{y}$$

Step 3: Apply Orthogonality Condition Replace $\frac{dy}{dx}$ with $-\frac{dx}{dy}$:

$$-\frac{dx}{dy} = 2\sqrt{y}$$

Step 4: Solve the New Differential Equation Rearrange to separate variables:

$$dx = -2\sqrt{y}\,dy$$

Integrate both sides:

$$\int dx = -2\int y^{1/2}\,dy$$
$$x = -2\left(\frac{y^{3/2}}{3/2}\right) + C$$
$$x = -2\left(\frac{2}{3}y^{3/2}\right) + C$$
$$x = -\frac{4}{3}y^{3/2} + C$$

Result: The orthogonal trajectories are the family of semi-cubical parabolas given by $3x + 4y^{3/2} = C'$.

Next Steps: Practice finding orthogonal trajectories with the Module 3 Practice Questions.

16 Module 3 Practice: Orthogonal Trajectories

This section contains practice problems for finding the geometric orthogonal trajectories of a given family of curves.

16.0.1 Question 1: Family of Circles

Find the orthogonal trajectories of: $x^2 + y^2 = k$

Solution:

1. **Differentiate to find the slope of the given family (m_1):** $2x + 2y\frac{dy}{dx} = 0$ $2y\frac{dy}{dx} = -2x \implies \frac{dy}{dx} = -\frac{x}{y}$ *(Note: The parameter k is already eliminated).*

2. **Apply Orthogonality Condition ($m_2 = -1/m_1$):** Replace $\frac{dy}{dx}$ with $-\frac{dx}{dy}$: $-\frac{dx}{dy} = -\frac{x}{y}$
3. **Solve the new DE:** $\frac{dx}{dy} = \frac{x}{y}$ Separate variables: $\frac{dx}{x} = \frac{dy}{y}$ Integrate: $\int \frac{1}{x} dx = \int \frac{1}{y} dy$ $\ln|x| = \ln|y| + \ln C$ $\ln|x| - \ln|y| = \ln C \implies \ln\left(\frac{x}{y}\right) = \ln C$ $\frac{x}{y} = C \implies x = Cy$ **Final Answer:** $y = C'x$ (A family of straight lines passing through the origin).

16.0.2 Question 2: Family of Parabolas

Find the orthogonal trajectories of: $y = cx^2$

Solution:

1. **Differentiate:** $\frac{dy}{dx} = 2cx$
2. **Eliminate parameter c:** From the original equation, $c = \frac{y}{x^2}$. Substitute c into the derivative: $\frac{dy}{dx} = 2\left(\frac{y}{x^2}\right)x = \frac{2y}{x}$
3. **Apply Orthogonality Condition:** Replace $\frac{dy}{dx}$ with $-\frac{dx}{dy}$: $-\frac{dx}{dy} = \frac{2y}{x}$
4. **Solve the new DE:** Separate variables: $-x\,dx = 2y\,dy$ Integrate: $-\int x\,dx = \int 2y\,dy$ $-\frac{x^2}{2} = y^2 + K$ Multiply by 2 to clear fractions: $-x^2 = 2y^2 + 2K$ Rearrange constants: **Final Answer:** $x^2 + 2y^2 = C$ (A family of ellipses).

16.0.3 Question 3: Exponential Curves

Find the orthogonal trajectories of: $y = ce^x$

Solution:

1. **Differentiate:** $\frac{dy}{dx} = ce^x$
2. **Eliminate parameter c:** Notice that ce^x is exactly equal to y. Substitute it directly: $\frac{dy}{dx} = y$
3. **Apply Orthogonality Condition:** Replace $\frac{dy}{dx}$ with $-\frac{dx}{dy}$: $-\frac{dx}{dy} = y$
4. **Solve the new DE:** Separate variables: $-dx = y\,dy$ Integrate: $-\int dx = \int y\,dy$ $-x = \frac{y^2}{2} + K$ Multiply by 2 and rearrange: $-2x = y^2 + 2K$ **Final Answer:** $2x + y^2 = C$ (A family of parabolas opening to the left).

Next Module: We now move to Higher-Order Differential Equations, starting with finding the Complementary Function. Proceed to Module 4: Higher-Order Linear ODEs.

17 Module 4: Higher-Order Linear ODEs (Constant Coefficients)

Welcome to **Module 4**. In the previous modules, we dealt with first-order equations where the highest derivative was $\frac{dy}{dx}$. Now, we advance to **Higher-Order Linear Differential Equations**, which involve second derivatives ($\frac{d^2y}{dx^2}$), third derivatives, and beyond.

These equations are fundamental in analyzing mechanical vibrations, electrical circuits (RLC circuits), and structural engineering. This module focuses specifically on linear equations with **Constant Coefficients**.

17.1 Module Learning Objectives

By the end of this module, you will be able to:

1. Understand the structure of the General Solution: $y = y_c + y_p$.
2. Solve homogeneous linear equations by finding the roots of the Characteristic Equation (to find y_c).
3. Apply specific methods to find the Particular Integral (y_p) for different types of non-homogeneous functions (Exponential, Algebraic, Trigonometric).

17.2 Topic Index & Solution Methods

The solution to a non-homogeneous linear ODE $f(D)y = F(x)$ is always the sum of two parts:

$$y = y_c + y_p$$

17.2.1 1. Finding the Complementary Function (y_c)

- **The Concept:** Solving the homogeneous part $f(D)y = 0$.
- **The Method:** Using the Auxiliary (Characteristic) Equation to find roots.
- **Cases:** Real & Distinct, Real & Repeated, and Complex Conjugate roots.

17.2.2 2. Particular Integral (y_p): Exponential Case

- **Function:** $F(x) = e^{ax}$
- **Rule:** Replace D with a.
- *Includes the "Case of Failure" where $f(a) = 0$.*

17.2.3 3. Particular Integral (y_p): Algebraic Case

- **Function:** $F(x) = x^n$ (Polynomials)
- **Rule:** Expand $[f(D)]^{-1}$ using the Binomial Theorem.
- *Includes step-by-step solved questions.*

17.2.4 4. Particular Integral (y_p): Trigonometric Case

- **Function:** $F(x) = \sin(ax)$ or $\cos(ax)$
- **Rule:** Replace D^2 with $-a^2$.
- *Includes the "Case of Failure" handling.*

17.2.5 5. Particular Integral (y_p): Shift & Product Rules

- **Function:** $F(x) = e^{ax}V(x)$ (Product of exponential and another function)
- **Rule:** The Exponential Shift Theorem.
- *Includes solved examples combining multiple methods.*

17.2.6 6. Particular Integral (y_p): Principle of Superposition

- **The Concept:** When $F(x)$ is a sum of different function types, split into parts, solve each separately, and add.

- **The Method:** Linearity of the differential operator allows solving each term independently.
- *Includes a step-by-step solved example.*

17.2.7 7. Practice Questions

- Comprehensive problems covering all y_c and y_p methods: homogeneous equations, exponential, algebraic, trigonometric, shift rule, superposition, and initial value problems.
- *Includes fully worked solutions.*

18 1. Finding the Complementary Function (y_c)

When solving higher-order linear differential equations of the form $f(D)y = F(x)$, the general solution is always composed of two parts:

$$y = y_c + y_p$$

The **Complementary Function** (y_c) represents the solution to the **homogeneous** version of the equation (where the right side is zero). This is the first step in solving *any* linear ODE with constant coefficients.

18.1 1. The Operator 'D'

To simplify writing derivatives, we use the differential operator D:

- $Dy = \frac{dy}{dx}$
- $D^2y = \frac{d^2y}{dx^2}$
- $D^ny = \frac{d^ny}{dx^n}$

An equation like $\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + 6y = 0$ can be written as:

$$(D^2 + 5D + 6)y = 0$$

18.2 2. The Auxiliary Equation (A.E.)

To find y_c, we assume a solution of the form $y = e^{mx}$. Substituting this into the homogeneous equation gives us the **Auxiliary Equation** (or Characteristic Equation).

Simply replace D with m and set the equation to zero:

$$f(m) = 0$$

We then solve for the roots of m. The nature of these roots determines the form of y_c.

18.3 3. Cases for Roots and Forms of y_c

18.3.1 Case 1: Real and Distinct Roots

If the roots $m_1, m_2, \ldots$ are real and different:

$$y_c = C_1e^{m_1x} + C_2e^{m_2x} + \ldots$$

18.3.2 Case 2: Real and Repeated Roots

If a root m is repeated k times (e.g., m, m, m):

$$y_c = (C_1 + C_2x + C_3x^2 + \cdots + C_kx^{k-1})e^{mx}$$

18.3.3 Case 3: Complex Conjugate Roots

If the roots are complex ($\alpha \pm i\beta$):

$$y_c = e^{\alpha x}(C_1 \cos \beta x + C_2 \sin \beta x)$$

18.4 4. Advanced: Finding Roots using Synthetic Division

When the auxiliary equation is a 3rd-degree or 4th-degree polynomial that cannot be easily factored, we use **Synthetic Division** to find the roots by trial and error.

18.4.1 Example: 4th-Order Equation

Find y_c for: $(D^4 + 6D^3 + 5D^2 - 24D - 36)y = 0$

Solution:

1. **Auxiliary Equation:** $m^4 + 6m^3 + 5m^2 - 24m - 36 = 0$
2. **First Root by Trial & Error:** Test small integers $(1, -1, 2, -2)$. Let's test $m = -2$: $(-2)^4 + 6(-2)^3 + 5(-2)^2 - 24(-2) - 36 = 16 - 48 + 20 + 48 - 36 = 0$. Since the result is 0, $m = -2$ **is a root**.
3. **Synthetic Division (First Pass):**

```
-2 |  1   6   5  -24  -36
   |     -2  -8    6   36
   -----------------------
      1   4  -3  -18    0  (Remainder is 0)
```

   The reduced cubic equation is: $m^3 + 4m^2 - 3m - 18 = 0$

4. **Second Root by Trial & Error:** Test $m = -2$ again on the new cubic equation: $(-2)^3 + 4(-2)^2 - 3(-2) - 18 = -8 + 16 + 6 - 18 = -4 \neq 0$. (Not a root). Test $m = 2$: $(2)^3 + 4(2)^2 - 3(2) - 18 = 8 + 16 - 6 - 18 = 0$. $m = 2$ **is a root.**
5. **Synthetic Division (Second Pass):**

```
 2 |  1   4  -3  -18
   |      2  12   18
   ------------------
      1   6   9    0
```

   The reduced quadratic equation is: $m^2 + 6m + 9 = 0$

6. **Solve the Quadratic:** $m^2 + 6m + 9 = 0 \implies (m + 3)^2 = 0 \implies m = -3, -3$
7. **Final Roots and y_c:** The four roots are: $m = -2, 2, -3, -3$. Since -3 is repeated, the Complementary Function is:

$$y_c = C_1e^{-2x} + C_2e^{2x} + (C_3 + C_4x)e^{-3x}$$

18.5 5. Solved Questions

Here are examples from the lecture notes demonstrating how to find y_c for different types of roots.

18.5.1 Question 1: Real & Distinct Roots

Find y_c for: $y'' + 6y' + 5y = 0$

Solution:

1. **Write in Operator Form:** $(D^2 + 6D + 5)y = 0$
2. **Auxiliary Equation:** $m^2 + 6m + 5 = 0$
3. **Solve for Roots:** Factor the quadratic: $(m+5)(m+1) = 0$ Roots: $m_1 = -1, m_2 = -5$
4. **Write y_c:** Since roots are real and distinct:

$$y_c = C_1 e^{-x} + C_2 e^{-5x}$$

18.5.2 Question 2: Real & Repeated Roots

Find y_c for: $y''' + 3y'' + 3y' + y = 0$

Solution:

1. **Write in Operator Form:** $(D^3 + 3D^2 + 3D + 1)y = 0$
2. **Auxiliary Equation:** $m^3 + 3m^2 + 3m + 1 = 0$
3. **Solve for Roots:** Recognize this as the binomial expansion of $(m+1)^3$: $(m+1)^3 = 0$ Roots: $m = -1, -1, -1$ (Repeated 3 times)
4. **Write y_c:** Since the root -1 repeats 3 times, we multiply the constants by increasing powers of x:

$$y_c = (C_1 + C_2 x + C_3 x^2)e^{-x}$$

18.5.3 Question 3: Complex Roots

Find y_c for: $y'''' + 4y = 0$

Solution:

1. **Write in Operator Form:** $(D^4 + 4)y = 0$

2. **Auxiliary Equation:** $m^4 + 4 = 0 \implies m^4 = -4$ $m^2 = \pm 2i$ $m = \pm\sqrt{2i}$

   Alternative Method (Factorization): $m^4 + 4 = (m^2+2)^2 - 4m^2 = (m^2+2-2m)(m^2+2+2m) = 0$

   Solve $m^2 - 2m + 2 = 0$ using quadratic formula: $m = \frac{2\pm\sqrt{4-8}}{2} = \frac{2\pm 2i}{2} = 1 \pm i$

   Solve $m^2 + 2m + 2 = 0$: $m = \frac{-2\pm\sqrt{4-8}}{2} = -1 \pm i$

   Roots: $1 \pm i$ and $-1 \pm i$ $(\alpha = 1, \beta = 1)$ and $(\alpha = -1, \beta = 1)$

3. **Write y_c:** Combine the solutions for both pairs of conjugate roots:

$$y_c = e^x(C_1 \cos x + C_2 \sin x) + e^{-x}(C_3 \cos x + C_4 \sin x)$$

Next Step: Once y_c is found, we must find the Particular Integral (y_p) based on the function on the right side of the equation. Proceed to Particular Integral: Exponential Case.

19 2. Particular Integral (y_p): Exponential Case

After finding the Complementary Function (y_c), the next step is to find the **Particular Integral** (y_p). This part of the solution depends entirely on the function $F(x)$ on the right-hand side of the differential equation.

This file covers **Method 1**, which is used when $F(x)$ is an exponential function of the form e^{ax}.

19.1 1. The Method

Given a differential equation $f(D)y = e^{ax}$:

The Particular Integral is written as:

$$y_p = \frac{1}{f(D)} e^{ax}$$

19.1.1 1.1 The Rule ($D \to a$)

To solve this, simply replace the differential operator D with the coefficient a from the exponent.

$$y_p = \frac{1}{f(a)} e^{ax}, \quad \text{provided } f(a) \neq 0$$

19.2 2. The Case of Failure ($f(a) = 0$)

If replacing D with a results in a zero in the denominator ($f(a) = 0$), the method fails. To fix this, we apply the following rule:

1. Multiply the numerator by x.
2. Differentiate the denominator with respect to D ($f'(D)$).
3. Try substituting $D = a$ again.

$$y_p = x \cdot \frac{1}{f'(D)} e^{ax}$$

If the denominator is still zero, repeat the process (multiply by x again to get x^2, and differentiate the denominator again to get $f''(D)$).

19.3 3. Solved Questions

Here are step-by-step examples from the lecture notes, covering both the standard case and the case of failure.

19.3.1 Question 1: Standard Case & Constants

Solve for y_p: $(D^2 - 6D + 9)y = 6e^{3x} + 7e^{-2x} - \ln 2$

Solution: We can split this into three separate parts for linearity:

$$y_p = y_{p1} + y_{p2} + y_{p3}$$

$$y_p = \frac{1}{(D-3)^2}(6e^{3x}) + \frac{1}{(D-3)^2}(7e^{-2x}) - \frac{1}{(D-3)^2}(\ln 2)$$

Part 1: $6e^{3x}$ **(**$a = 3$**)** Substitute $D = 3$ into $(D-3)^2$: $(3-3)^2 = 0$. **Case of Failure!**

- Multiply by x, differentiate denominator $2(D-3)$:

$$x \cdot \frac{1}{2(D-3)} 6e^{3x}$$

- Substitute $D = 3$ again: $2(3-3) = 0$. **Failure again!**
- Multiply by x again (x^2), differentiate denominator again (2):

$$x^2 \cdot \frac{1}{2} 6e^{3x} = 3x^2 e^{3x}$$

Part 2: $7e^{-2x}$ **(**$a = -2$**)** Substitute $D = -2$:

$$\frac{1}{(-2-3)^2} 7e^{-2x} = \frac{1}{(-5)^2} 7e^{-2x} = \frac{7}{25} e^{-2x}$$

Part 3: Constant Term $\ln 2$ **(**$a = 0$**)** A constant can be written as $(\ln 2) \cdot e^{0x}$. Here, $a = 0$. Substitute $D = 0$:

$$\frac{1}{(0-3)^2}(\ln 2) = \frac{1}{9} \ln 2$$

Total y_p:

$$y_p = 3x^2 e^{3x} + \frac{7}{25} e^{-2x} - \frac{\ln 2}{9}$$

19.3.2 Question 2: Higher Order Failure

Solve for y_p: $(D^3 - 3D^2 + 3D - 1)y = e^x$

Solution:

1. **Identify Form:** The operator $(D^3 - 3D^2 + 3D - 1)$ is the expansion of $(D-1)^3$.

$$y_p = \frac{1}{(D-1)^3} e^x$$

2. **Apply Rule ($D \to 1$):** Substitute $D = 1$: $(1-1)^3 = 0$. **Case of Failure.**
3. **Fix 1:** Multiply by x, differentiate denominator ($3(D-1)^2$).

$$x \cdot \frac{1}{3(D-1)^2} e^x$$

   Substitute $D = 1$: Still 0.
4. **Fix 2:** Multiply by x again (x^2), differentiate denominator ($6(D-1)$).

$$x^2 \cdot \frac{1}{6(D-1)} e^x$$

   Substitute $D = 1$: Still 0.
5. **Fix 3:** Multiply by x again (x^3), differentiate denominator (6).

$$x^3 \cdot \frac{1}{6} e^x$$

   Now the denominator is constant (6), which is not zero.

Final Answer:

$$y_p = \frac{x^3 e^x}{6}$$

Next Method: What if $F(x)$ is not exponential, but a polynomial like x^2 or x^3? Proceed to Particular Integral: Algebraic Case.

20 3. Particular Integral (y_p): Algebraic Case

When the function on the right side of the differential equation, $F(x)$, is an algebraic polynomial (like $x, x^2, 1+2x, x^4$, etc.), we cannot use simple substitution. Instead, we use the **Binomial Expansion** method.

20.1 1. The Method (Binomial Expansion)

Given a differential equation $f(D)y = x^n$:

$$y_p = \frac{1}{f(D)} x^n$$

To solve this, we must expand the operator $\frac{1}{f(D)}$ into an infinite series of derivatives ($1 + AD + BD^2 + \ldots$). Since the derivative of a polynomial x^n eventually becomes zero (e.g., $D^{n+1}(x^n) = 0$), the series will terminate, giving us a finite answer.

20.1.1 1.1 The Working Rule

1. **Factor out the lowest degree term** from the denominator $f(D)$ to create a term of the form $(1 \pm \phi(D))$.
2. **Bring the term to the numerator** with a negative power: $(1 \pm \phi(D))^{-1}$.
3. **Expand** using the Binomial Theorem.
4. **Operate** the derivatives on x^n.

20.1.2 1.2 Important Binomial Formulas

- $(1+x)^{-1} = 1 - x + x^2 - x^3 + \dots$
- $(1-x)^{-1} = 1 + x + x^2 + x^3 + \dots$
- $(1+x)^{-2} = 1 - 2x + 3x^2 - \dots$

20.2 2. Solved Questions

Here are three step-by-step examples from the lecture notes, ranging from simple linear polynomials to higher-order equations involving integration.

20.2.1 Question 1: Linear Polynomial

Solve for y_p: $(D^2 + 5D + 4)y = 3 - 2x$

Solution:

$$y_p = \frac{1}{D^2 + 5D + 4}(3 - 2x)$$

Step 1: Factor out the lowest degree term (constant 4)

$$y_p = \frac{1}{4\left(1 + \frac{5D+D^2}{4}\right)}(3 - 2x)$$

Step 2: Bring to numerator and Expand Since $(3 - 2x)$ is a polynomial of degree 1, we only need to expand up to D^1. Higher powers (D^2, D^3) applied to x will be zero.

$$y_p = \frac{1}{4}\left[1 + \frac{5D}{4}\right]^{-1}(3 - 2x)$$

Using $(1 + x)^{-1} \approx 1 - x$:

$$y_p = \frac{1}{4}\left[1 - \frac{5D}{4}\right](3 - 2x)$$

Step 3: Operate

$$y_p = \frac{1}{4}\left[1(3 - 2x) - \frac{5}{4}D(3 - 2x)\right]$$

Differentiate $(3 - 2x)$: $D(3 - 2x) = -2$.

$$y_p = \frac{1}{4}\left[(3 - 2x) - \frac{5}{4}(-2)\right]$$

$$y_p = \frac{1}{4}\left[3 - 2x + \frac{5}{2}\right]$$

$$y_p = \frac{1}{4}\left[\frac{6 - 4x + 5}{2}\right] = \frac{11 - 4x}{8}$$

20.2.2 Question 2: Inverse Operators (Integration)

Solve for y_p: $D^2(D^2+4D)y = 96x^2$ *(Note: The operator simplifies to $D^3(D+4)$).*

Solution:

$$y_p = \frac{1}{D^3(D+4)}96x^2$$

Step 1: Factor out lowest term from $(D+4)$ Factor out 4:

$$y_p = \frac{1}{D^3 \cdot 4(1+D/4)}96x^2$$

$$y_p = \frac{24}{D^3}\left(1+\frac{D}{4}\right)^{-1}x^2$$

Step 2: Expand We need to expand up to D^2 because we are operating on x^2.

$$y_p = \frac{24}{D^3}\left[1-\frac{D}{4}+\left(\frac{D}{4}\right)^2-\dots\right]x^2$$

$$y_p = \frac{24}{D^3}\left[x^2-\frac{1}{4}D(x^2)+\frac{1}{16}D^2(x^2)\right]$$

Step 3: Derivatives

- $D(x^2) = 2x$
- $D^2(x^2) = 2$

$$y_p = \frac{24}{D^3}\left[x^2-\frac{1}{4}(2x)+\frac{1}{16}(2)\right]$$

$$y_p = \frac{24}{D^3}\left[x^2-\frac{x}{2}+\frac{1}{8}\right]$$

Step 4: Integrate $(1/D^3)$ The operator $\frac{1}{D^3}$ means "integrate 3 times". However, the notes use a shortcut by keeping $\frac{1}{D}$ terms inside. Let's follow the standard integration path.

1. Integrate $x^2 \to \frac{x^5}{60}$ (3 times)
2. Integrate $x \to \frac{x^4}{24}$ (3 times)
3. Integrate const $\to \frac{x^3}{6}$ (3 times)

Alternative Interpretation from Notes: The notes solve it as $D^2(D^2+4D) \to \frac{24}{D}[x^3/3 - x^2/4] \to 2x^4 - 6x^2$. Let's re-evaluate based on the specific structure in the notes image:

$$y_p = 2x^4 - 6x^2$$

20.2.3 Question 3: Higher Powers

Solve for y_p: $(D^4 + 4)y = x^4$

Solution:

$$y_p = \frac{1}{D^4 + 4} x^4$$

Step 1: Factor out 4

$$y_p = \frac{1}{4(1 + D^4/4)} x^4 = \frac{1}{4}\left(1 + \frac{D^4}{4}\right)^{-1} x^4$$

Step 2: Expand

$$y_p = \frac{1}{4}\left[1 - \frac{D^4}{4}\right] x^4$$

Step 3: Operate

$$y_p = \frac{1}{4}\left[x^4 - \frac{1}{4}D^4(x^4)\right]$$

Calculate $D^4(x^4)$: $x^4 \xrightarrow{D} 4x^3 \xrightarrow{D} 12x^2 \xrightarrow{D} 24x \xrightarrow{D} 24$

$$y_p = \frac{1}{4}\left[x^4 - \frac{1}{4}(24)\right]$$

$$y_p = \frac{1}{4}(x^4 - 6)$$

$$y_p = \frac{x^4}{4} - \frac{3}{2}$$

Next Method: How do we handle sine and cosine functions? Proceed to Particular Integral: Trigonometric Case.

21 4. Particular Integral (y_p): Trigonometric Case

When the right-hand side of a linear differential equation contains sine or cosine functions ($F(x) = \sin ax$ or $\cos ax$), we use a specific substitution rule involving the square of the differential operator (D^2).

21.1 1. The Method ($D^2 \to -a^2$)

Given a differential equation $f(D^2)y = \sin ax$ (or $\cos ax$):

The Particular Integral is:

$$y_p = \frac{1}{f(D^2)} \sin ax$$

21.1.1 1.1 The Rule

Replace every instance of D^2 with $-a^2$.

$$y_p = \frac{1}{f(-a^2)} \sin ax$$

Important Warning: The negative sign is **outside** the square.

- If $a = 2$, then D^2 becomes $-(2^2) = -4$.
- It does **NOT** become $(-2)^2 = +4$.

21.1.2 1.2 Handling Remaining D Terms (Rationalization)

If replacing D^2 leaves a D term in the denominator (e.g., $D - 3$), you cannot substitute $-a^2$ for D directly. Instead:

1. Multiply the numerator and denominator by the conjugate (e.g., $D + 3$).
2. This creates a difference of squares ($D^2 - 9$) in the denominator.
3. Replace the new D^2 with $-a^2$ again.
4. Apply the numerator operator to the function.

21.2 2. The Case of Failure ($f(-a^2) = 0$)

If replacing D^2 with $-a^2$ causes the denominator to become zero, the method fails. We apply the standard failure rule:

1. Multiply the numerator by x.
2. Differentiate the denominator with respect to D ($f'(D)$).
3. Try the substitution again.

$$y_p = x \cdot \frac{1}{f'(D)} \sin ax$$

21.3 3. Solved Questions

Here are three distinct cases from the lecture notes: a standard substitution, a rationalization case, and a case of failure.

21.3.1 Question 1: Standard Substitution

Solve for y_p: $(D^2 + 4)y = \sin 3x$

Solution:

$$y_p = \frac{1}{D^2 + 4} \sin 3x$$

Step 1: Identify a Here, $a = 3$. Therefore, $-a^2 = -(3^2) = -9$.

Step 2: Substitute $D^2 = -9$

$$y_p = \frac{1}{-9 + 4} \sin 3x$$

$$y_p = \frac{1}{-5}\sin 3x$$

Final Answer:

$$y_p = -\frac{1}{5}\sin 3x$$

21.3.2 Question 2: Rationalization Required

Solve for y_p: $(D^2 + D + 1)y = \cos 2x$

Solution:

$$y_p = \frac{1}{D^2 + D + 1}\cos 2x$$

Step 1: Identify a and Substitute Here, $a = 2$. So, $D^2 \to -4$.

$$y_p = \frac{1}{-4 + D + 1}\cos 2x$$

$$y_p = \frac{1}{D - 3}\cos 2x$$

Step 2: Rationalize the Denominator We still have a D in the denominator. Multiply numerator and denominator by the conjugate $(D + 3)$.

$$y_p = \frac{D + 3}{(D - 3)(D + 3)}\cos 2x$$

$$y_p = \frac{D + 3}{D^2 - 9}\cos 2x$$

Step 3: Substitute D^2 again Replace D^2 with -4 again.

$$y_p = \frac{D + 3}{-4 - 9}\cos 2x$$

$$y_p = \frac{D + 3}{-13}\cos 2x$$

Step 4: Operate Numerator Apply $(D + 3)$ to $\cos 2x$:

$$y_p = -\frac{1}{13}[D(\cos 2x) + 3\cos 2x]$$

- Derivative $D(\cos 2x) = -2\sin 2x$

$$y_p = -\frac{1}{13}[-2\sin 2x + 3\cos 2x]$$

Final Answer:

$$y_p = \frac{2\sin 2x - 3\cos 2x}{13}$$

21.3.3 Question 3: Case of Failure

Solve for y_p: $(D^2 + 4)y = \cos 2x$

Solution:

$$y_p = \frac{1}{D^2 + 4} \cos 2x$$

Step 1: Check Substitution Here $a = 2$, so $D^2 \to -4$. Denominator becomes $-4+4 = 0$. **Method Fails.**

Step 2: Apply Failure Rule Multiply by x and differentiate the denominator ($D^2+4 \to 2D$).

$$y_p = x \cdot \frac{1}{2D} \cos 2x$$

Step 3: Interpret $1/D$ The operator $\frac{1}{D}$ represents **Integration**.

$$y_p = \frac{x}{2} \int \cos 2x \, dx$$

Step 4: Integrate

$$\int \cos 2x \, dx = \frac{\sin 2x}{2}$$

$$y_p = \frac{x}{2} \left(\frac{\sin 2x}{2} \right)$$

Final Answer:

$$y_p = \frac{x \sin 2x}{4}$$

Next Method: Sometimes we have a product of two functions, like $e^x \sin x$. For this, we use the Shift Rule. Proceed to Exponential Shift Rule.

22 5. Particular Integral (y_p): Exponential Shift Rule

Often, the non-homogeneous term $F(x)$ on the right side of a differential equation is not a single function, but a **product** of an exponential function and another function (like a polynomial or a trigonometric function).

To solve this, we use a powerful technique known as the **Exponential Shift Theorem**.

22.1 1. The Method (Exponential Shift Theorem)

Given a differential equation $f(D)y = e^{ax}V(x)$, where $V(x)$ is any other function (usually $\sin bx$, $\cos bx$, or x^n).

The Particular Integral is:

$$y_p = \frac{1}{f(D)}[e^{ax}V(x)]$$

22.1.1 1.1 The Working Rule ($D \to D + a$)

You cannot simply substitute $D = a$ because $V(x)$ is also present. Instead, you "shift" the exponential function to the front of the operator by modifying the operator itself.

1. **Shift:** Move e^{ax} to the left of the fraction.
2. **Replace:** Replace every D in the denominator with $(D + a)$.
3. **Simplify:** Expand and simplify the new denominator $f(D + a)$.
4. **Operate:** You are now left with $e^{ax} \cdot \frac{1}{f(D+a)} V(x)$. Apply the standard rules for $V(x)$ (Trigonometric rule if $V(x)$ is sin / cos, or Binomial rule if $V(x)$ is x^n) to finish the problem.

$$y_p = e^{ax} \frac{1}{f(D + a)} V(x)$$

22.2 2. Solved Questions

Here are two detailed examples from the lecture notes demonstrating how this shift reduces complex products into standard problems we already know how to solve.

22.2.1 Question 1: Product with Trigonometric Function

Solve for y_p: $(D^2 + 2D + 4)y = e^x \sin 2x$

Solution:

$$y_p = \frac{1}{D^2 + 2D + 4}(e^x \sin 2x)$$

Step 1: Apply Exponential Shift Here, the exponential is e^{1x}, so $a = 1$. We move e^x to the front and replace every D with $(D + 1)$.

$$y_p = e^x \frac{1}{(D + 1)^2 + 2(D + 1) + 4} \sin 2x$$

Step 2: Simplify the Denominator

$$y_p = e^x \frac{1}{(D^2 + 2D + 1) + 2D + 2 + 4} \sin 2x$$

$$y_p = e^x \frac{1}{D^2 + 4D + 7} \sin 2x$$

Step 3: Apply Trigonometric Rule to $V(x)$ Now we just solve for $\sin 2x$. The rule is to replace D^2 with $-b^2 = -(2^2) = -4$.

$$y_p = e^x \frac{1}{-4 + 4D + 7} \sin 2x$$

$$y_p = e^x \frac{1}{4D + 3} \sin 2x$$

Step 4: Rationalize and Finalize Multiply the numerator and denominator by the conjugate $(4D - 3)$:

$$y_p = e^x \frac{4D - 3}{(4D + 3)(4D - 3)} \sin 2x$$

$$y_p = e^x \frac{4D-3}{16D^2-9} \sin 2x$$

Replace D^2 with -4 again:

$$y_p = e^x \frac{4D-3}{16(-4)-9} \sin 2x = e^x \frac{4D-3}{-64-9} \sin 2x$$

$$y_p = e^x \frac{4D-3}{-73} \sin 2x$$

Apply the numerator operator $(4D-3)$ to $\sin 2x$:

- $4D(\sin 2x) = 4(2\cos 2x) = 8\cos 2x$
- $-3(\sin 2x) = -3\sin 2x$

Final Answer:

$$y_p = -\frac{e^x}{73}(8\cos 2x - 3\sin 2x)$$

22.2.2 Question 2: Shift Rule resulting in a Case of Failure

Solve for y_p: $(D^2 - 4D + 13)y = e^{2x}\cos 3x$

Solution:

$$y_p = \frac{1}{D^2 - 4D + 13}(e^{2x}\cos 3x)$$

Step 1: Apply Exponential Shift Here, $a = 2$. Move e^{2x} to the front and replace D with $(D+2)$.

$$y_p = e^{2x} \frac{1}{(D+2)^2 - 4(D+2) + 13} \cos 3x$$

Step 2: Simplify the Denominator

$$y_p = e^{2x} \frac{1}{(D^2 + 4D + 4) - 4D - 8 + 13} \cos 3x$$

$$y_p = e^{2x} \frac{1}{D^2 + 9} \cos 3x$$

Step 3: Apply Trigonometric Rule to $V(x)$ For $\cos 3x$, replace D^2 with $-b^2 = -(3^2) = -9$.

$$\text{Denominator } = -9 + 9 = 0$$

This is a Case of Failure!

Step 4: Apply Failure Rule Multiply the numerator by x and differentiate the denominator with respect to D. The denominator $D^2 + 9$ differentiates to $2D$.

$$y_p = e^{2x} \cdot x \cdot \frac{1}{2D} \cos 3x$$

Step 5: Integrate $(1/D)$ The operator $\frac{1}{D}$ means integration.

$$y_p = \frac{xe^{2x}}{2} \int \cos 3x \, dx$$

$$\int \cos 3x \, dx = \frac{\sin 3x}{3}$$

Final Answer:

$$y_p = \frac{xe^{2x}}{2}\left(\frac{\sin 3x}{3}\right) = \frac{xe^{2x}\sin 3x}{6}$$

Next Method: What if $F(x)$ is a *sum* of different function types? We can split the problem into parts. Proceed to Principle of Superposition.

23 6. Principle of Superposition (Sum of Functions)

When the function $F(x)$ on the right side is a sum of different types of functions (e.g., an exponential plus a trigonometric function), we apply the **Principle of Superposition**. We split the Particular Integral into separate parts, solve each using its specific rule, and add them together.

23.0.1 Example: Mixed Functions

Solve: $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} - 3y = 2e^{2x} + 10\sin 3x$

Solution: **1. Find** y_c**:** $m^2 - 2m - 3 = 0 \implies (m-3)(m+1) = 0 \implies m = 3, -1$ $y_c = C_1e^{3x} + C_2e^{-x}$

2. Set up y_p**:** $y_p = \frac{1}{D^2-2D-3}(2e^{2x} + 10\sin 3x)$ Split into two parts: $y_p = y_{p1} + y_{p2}$

3. Solve y_{p1} **(Exponential Rule):** $y_{p1} = 2\frac{1}{D^2-2D-3}e^{2x}$ Substitute $D = 2$: $y_{p1} = 2\frac{1}{(2)^2-2(2)-3}e^{2x} = 2\frac{1}{4-4-3}e^{2x} = -\frac{2}{3}e^{2x}$

4. Solve y_{p2} **(Trigonometric Rule):** $y_{p2} = 10\frac{1}{D^2-2D-3}\sin 3x$ Substitute $D^2 = -(3^2) = -9$: $y_{p2} = 10\frac{1}{-9-2D-3}\sin 3x = 10\frac{1}{-2D-12}\sin 3x = -5\frac{1}{D+6}\sin 3x$ Rationalize the denominator by multiplying by $(D-6)$: $y_{p2} = -5\frac{D-6}{D^2-36}\sin 3x$ Substitute $D^2 = -9$ again: $y_{p2} = -5\frac{D-6}{-9-36}\sin 3x = \frac{-5}{-45}(D-6)\sin 3x = \frac{1}{9}[D(\sin 3x) - 6\sin 3x]$ $y_{p2} = \frac{1}{9}(3\cos 3x - 6\sin 3x) = \frac{1}{3}\cos 3x - \frac{2}{3}\sin 3x$

5. Final General Solution: $y = y_c + y_{p1} + y_{p2}$

$$y = C_1e^{3x} + C_2e^{-x} - \frac{2}{3}e^{2x} + \frac{1}{3}\cos 3x - \frac{2}{3}\sin 3x$$

Next Steps: Put all the methods together with the Module 4 Practice Questions.

24 Module 4 Practice: Higher-Order Linear ODEs

This section contains all the comprehensive practice problems and homework questions for solving linear differential equations with constant coefficients. Remember that the general solution is always $y = y_c + y_p$.

24.1 Part A: Homogeneous Equations (Finding y_c only)

For these problems, $F(x) = 0$, so $y_p = 0$ and the general solution is just $y = y_c$.

24.1.1 Question 1

Solve: $(D^4 - D^3 - 9D^2 - 11D - 4)y = 0$

Solution:

1. **Auxiliary Equation:** $m^4 - m^3 - 9m^2 - 11m - 4 = 0$
2. **Find roots using Synthetic Division:** Test $m = -1$: $1 - (-1) - 9(-1)^2 - 11(-1) - 4 = 1 + 1 - 9 + 11 - 4 = 0$. So $m = -1$ is a root.

```
-1 |  1   -1   -9   -11   -4
   |      -1    2     7    4
   ---------------------------
      1   -2   -7    -4    0
```

The reduced cubic is $m^3 - 2m^2 - 7m - 4 = 0$. Test $m = -1$ again: $-1 - 2(1) - 7(-1) - 4 = -1 - 2 + 7 - 4 = 0$.

```
-1 |  1   -2   -7   -4
   |      -1    3    4
   ---------------------
      1   -3   -4    0
```

The reduced quadratic is $m^2 - 3m - 4 = 0$.

3. **Solve Quadratic:** $(m - 4)(m + 1) = 0 \implies m = 4, m = -1$.
4. **Final Roots:** $-1, -1, -1, 4$.
5. **General Solution:**

$$y = (C_1 + C_2x + C_3x^2)e^{-x} + C_4e^{4x}$$

24.1.2 Question 2

Solve: $(D^4 + 5D^2 - 36)y = 0$

Solution:

1. **Auxiliary Equation:** $m^4 + 5m^2 - 36 = 0$
2. **Factorize:** Let $k = m^2$. Then $k^2 + 5k - 36 = 0$. $(k + 9)(k - 4) = 0 \implies (m^2 + 9)(m^2 - 4) = 0$.
3. **Find roots:** $m^2 = -9 \implies m = \pm 3i$ $m^2 = 4 \implies m = \pm 2$
4. **General Solution:**

$$y = C_1e^{2x} + C_2e^{-2x} + C_3 \cos 3x + C_4 \sin 3x$$

24.1.3 Question 3

Solve: $(D^2 - 2D + 5)^2y = 0$

Solution:

1. **Auxiliary Equation:** $(m^2 - 2m + 5)^2 = 0$

2. **Find roots of the inner quadratic:** $m = \frac{-(-2)\pm\sqrt{(-2)^2-4(1)(5)}}{2(1)} = \frac{2\pm\sqrt{4-20}}{2} = \frac{2\pm 4i}{2} = 1 \pm 2i$.
3. **Handle squared term:** Because the entire quadratic is squared, the complex conjugate roots are repeated twice: $1 \pm 2i, 1 \pm 2i$.
4. **General Solution:**

$$y = e^x \left[(C_1 + C_2x)\cos 2x + (C_3 + C_4x)\sin 2x\right]$$

24.1.4 Question 4

Solve: $(D^2 + 25)y = 0$

Solution:

1. **Auxiliary Equation:** $m^2 + 25 = 0 \implies m^2 = -25 \implies m = \pm 5i$.
2. **General Solution:**

$$y = C_1 \cos 5x + C_2 \sin 5x$$

24.1.5 Question 5

Solve: $(D^4 + 4D^2)y = 0$

Solution:

1. **Auxiliary Equation:** $m^4 + 4m^2 = 0 \implies m^2(m^2 + 4) = 0$.
2. **Find roots:** $m^2 = 0 \implies m = 0, 0$ $m^2 = -4 \implies m = \pm 2i$
3. **General Solution:**

$$y = (C_1 + C_2x)e^{0x} + C_3 \cos 2x + C_4 \sin 2x = C_1 + C_2x + C_3 \cos 2x + C_4 \sin 2x$$

24.1.6 Question 6

Solve: $\frac{d^5y}{dx^5} + 12\frac{d^4y}{dx^4} + 36\frac{d^3y}{dx^3} = 0$

Solution:

1. **Auxiliary Equation:** $m^5 + 12m^4 + 36m^3 = 0$.
2. **Factorize:** $m^3(m^2 + 12m + 36) = 0 \implies m^3(m + 6)^2 = 0$.
3. **Find roots:** $m = 0, 0, 0, -6, -6$.
4. **General Solution:**

$$y = C_1 + C_2x + C_3x^2 + (C_4 + C_5x)e^{-6x}$$

24.1.7 Question 7

Solve: $\frac{d^5y}{dx^5} + 5\frac{d^4y}{dx^4} - 2\frac{d^3y}{dx^3} - 10\frac{d^2y}{dx^2} + \frac{dy}{dx} + 5y = 0$

Solution:

1. **Auxiliary Equation:** $m^5 + 5m^4 - 2m^3 - 10m^2 + m + 5 = 0$.
2. **Factor by grouping:** $m^4(m+5) - 2m^2(m+5) + 1(m+5) = 0$ $(m^4 - 2m^2 + 1)(m+5) = 0$ $(m^2-1)^2(m+5) = 0$ $((m-1)(m+1))^2(m+5) = 0 \implies (m-1)^2(m+1)^2(m+5) = 0$.
3. **Find roots:** $m = 1, 1, -1, -1, -5$.

4. **General Solution:**

$$y = (C_1 + C_2x)e^x + (C_3 + C_4x)e^{-x} + C_5e^{-5x}$$

24.2 Part B: Particular Integral - Exponential Case

24.2.1 Question 8 (Standard Case)

Solve: $\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = e^{3x}$

Solution:

1. **Find y_c:** $m^2-3m+2 = 0 \implies (m-1)(m-2) = 0 \implies m = 1, 2$. $y_c = C_1e^x+C_2e^{2x}$.
2. **Find y_p:** $y_p = \frac{1}{D^2-3D+2}e^{3x}$. Substitute $D = 3$: $y_p = \frac{1}{9-9+2}e^{3x} = \frac{1}{2}e^{3x}$.
3. **General Solution:**
$$y = C_1e^x + C_2e^{2x} + \frac{1}{2}e^{3x}$$

24.2.2 Question 9 (Case of Failure)

Solve: $(D^3 + 2D^2 - D - 2)y = e^x$

Solution:

1. **Find y_c:** $m^3+2m^2-m-2 = 0 \implies m^2(m+2)-1(m+2) = 0 \implies (m^2-1)(m+2) = 0$. Roots: $1, -1, -2$. $y_c = C_1e^x + C_2e^{-x} + C_3e^{-2x}$.
2. **Find y_p:** $y_p = \frac{1}{D^3+2D^2-D-2}e^x$. Substitute $D = 1$: Denominator becomes $1 + 2 - 1 - 2 = 0$. Failure. Multiply by x, diff denominator: $y_p = x\frac{1}{3D^2+4D-1}e^x$. Substitute $D = 1$: $y_p = x\frac{1}{3(1)+4(1)-1}e^x = \frac{x}{6}e^x$.
3. **General Solution:**
$$y = C_1e^x + C_2e^{-x} + C_3e^{-2x} + \frac{x}{6}e^x$$

24.2.3 Question 10 (Case of Failure)

Solve: $(D^3 - 2D^2 - 5D + 6)y = e^{3x}$

Solution:

1. **Find y_c:** $m^3 - 2m^2 - 5m + 6 = 0$. By synthetic division (root 1 works), factors to $(m - 1)(m + 2)(m - 3) = 0$. Roots: $1, -2, 3$. $y_c = C_1e^x + C_2e^{-2x} + C_3e^{3x}$.
2. **Find y_p:** $y_p = \frac{1}{D^3-2D^2-5D+6}e^{3x}$. Substitute $D = 3$: Denom $= 27 - 18 - 15 + 6 = 0$. Failure. Multiply by x, diff denominator: $y_p = x\frac{1}{3D^2-4D-5}e^{3x}$. Substitute $D = 3$: $y_p = x\frac{1}{27-12-5}e^{3x} = \frac{x}{10}e^{3x}$.
3. **General Solution:**
$$y = C_1e^x + C_2e^{-2x} + C_3e^{3x} + \frac{x}{10}e^{3x}$$

24.2.4 Question 11 (Case of Failure with Complex Roots)

Solve: $\frac{d^3y}{dx^3} - \frac{d^2y}{dx^2} + 4\frac{dy}{dx} - 4y = e^x$

Solution:

1. **Find y_c:** $m^3-m^2+4m-4=0 \implies m^2(m-1)+4(m-1)=0 \implies (m^2+4)(m-1)=0$. Roots: $1, \pm 2i$. $y_c = C_1e^x + C_2\cos 2x + C_3 \sin 2x$.
2. **Find y_p:** $y_p = \frac{1}{D^3-D^2+4D-4}e^x$. Substitute $D=1$: Denom $= 1-1+4-4=0$. Failure. Multiply by x, diff denominator: $y_p = x\frac{1}{3D^2-2D+4}e^x$. Substitute $D=1$: $y_p = x\frac{1}{3-2+4}e^x = \frac{x}{5}e^x$.
3. **General Solution:**

$$y = C_1e^x + C_2\cos 2x + C_3\sin 2x + \frac{x}{5}e^x$$

24.2.5 Question 12 (Triple Failure)

Solve: $\frac{d^3y}{dx^3} + 3\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + y = e^{-x}$

Solution:

1. **Find y_c:** $m^3+3m^2+3m+1=0 \implies (m+1)^3=0$. Roots: $-1,-1,-1$. $y_c = (C_1+C_2x+C_3x^2)e^{-x}$.
2. **Find y_p:** $y_p = \frac{1}{(D+1)^3}e^{-x}$. Sub $D=-1$ fails. Diff 1: $x\frac{1}{3(D+1)^2}e^{-x}$ (fails). Diff 2: $x^2\frac{1}{6(D+1)}e^{-x}$ (fails). Diff 3: $x^3\frac{1}{6}e^{-x}$.
3. **General Solution:**

$$y = (C_1+C_2x+C_3x^2)e^{-x} + \frac{x^3}{6}e^{-x}$$

24.2.6 Question 13 (Superposition of Exponentials)

Solve: $(D-2)(D+1)^2y = e^{2x}+e^{-x}$

Solution:

1. **Find y_c:** Roots are $2,-1,-1$. $y_c = C_1e^{2x} + (C_2+C_3x)e^{-x}$.
2. **Find y_p:** Split into $y_{p1}+y_{p2}$. $y_{p1} = \frac{1}{(D-2)(D+1)^2}e^{2x}$. Sub $D=2$ fails. Diff denominator product rule, or just multiply by x and drop the $(D-2)$ factor: $x\frac{1}{1\cdot(2+1)^2}e^{2x} = \frac{x}{9}e^{2x}$. $y_{p2} = \frac{1}{(D-2)(D+1)^2}e^{-x}$. Sub $D=-1$ fails twice. Multiply by x^2 and drop the $(D+1)^2$ factor: $x^2\frac{1}{2!(-1-2)}e^{-x} = -\frac{x^2}{6}e^{-x}$.
3. **General Solution:**

$$y = C_1e^{2x} + (C_2+C_3x)e^{-x} + \frac{x}{9}e^{2x} - \frac{x^2}{6}e^{-x}$$

24.2.7 Question 14 (Standard Case)

Solve: $(D-1)^3y = 16e^{3x}$

Solution:

1. **Find y_c:** Roots $1,1,1$. $y_c = (C_1+C_2x+C_3x^2)e^x$.
2. **Find y_p:** $y_p = \frac{1}{(D-1)^3}16e^{3x} = 16\frac{1}{(3-1)^3}e^{3x} = 16\frac{1}{8}e^{3x} = 2e^{3x}$.
3. **General Solution:**

$$y = (C_1+C_2x+C_3x^2)e^x + 2e^{3x}$$

24.3 Part C: Particular Integral - Algebraic Case

24.3.1 Question 15

Solve: $(D^2+5D+4)y = 3-2x$

Solution:

1. **Find** y_c**:** $m^2+5m+4=0 \implies m=-1,-4$. $y_c = C_1e^{-x}+C_2e^{-4x}$.
2. **Find** y_p**:** $y_p = \frac{1}{4(1+\frac{5D+D^2}{4})}(3-2x)$ Expand: $y_p = \frac{1}{4}\left[1-\frac{5D}{4}\right](3-2x)$ Operate: $y_p = \frac{1}{4}\left[(3-2x)-\frac{5}{4}(-2)\right] = \frac{1}{4}\left[3-2x+\frac{5}{2}\right] = \frac{11-4x}{8}$.
3. **General Solution:**
$$y = C_1e^{-x}+C_2e^{-4x}+\frac{11-4x}{8}$$

24.3.2 Question 16

Solve: $\frac{d^3y}{dx^3}-\frac{d^2y}{dx^2}-6\frac{dy}{dx} = 1+x^2$

Solution:

1. **Find** y_c**:** $m(m^2-m-6)=0 \implies m(m-3)(m+2)=0$. Roots: $0,3,-2$. $y_c = C_1+C_2e^{3x}+C_3e^{-2x}$.
2. **Find** y_p**:** $y_p = \frac{1}{-6D(1+\frac{D^2-D}{-6})}(1+x^2) = -\frac{1}{6D}\left[1-\left(\frac{D-D^2}{6}\right)\right]^{-1}(1+x^2)$. Expand to D^2: $-\frac{1}{6D}\left[1+\frac{D-D^2}{6}+\frac{D^2}{36}\right](1+x^2) = -\frac{1}{6D}\left[1+\frac{D}{6}-\frac{5D^2}{36}\right](1+x^2)$ Operate: $-\frac{1}{6D}\left[(1+x^2)+\frac{1}{6}(2x)-\frac{5}{36}\right.$ $-\frac{1}{6D}\left[x^2+\frac{x}{3}+\frac{13}{18}\right]$. Integrate $(1/D)$: $-\frac{1}{6}\left(\frac{x^3}{3}+\frac{x^2}{6}+\frac{13x}{18}\right)$.
3. **General Solution:**
$$y = C_1+C_2e^{3x}+C_3e^{-2x}-\frac{x^3}{18}-\frac{x^2}{36}-\frac{13x}{108}$$

24.3.3 Question 17

Solve: $\frac{d^4y}{dx^4}+4y = x^4$

Solution:

1. **Find** y_c**:** $m^4+4=0 \implies$ roots $1\pm i, -1\pm i$. $y_c = e^x(C_1\cos x+C_2\sin x)+e^{-x}(C_3\cos x+C_4\sin x)$.
2. **Find** y_p**:** $y_p = \frac{1}{4(1+D^4/4)}x^4 = \frac{1}{4}(1-D^4/4)x^4 = \frac{1}{4}(x^4-\frac{24}{4}) = \frac{x^4}{4}-\frac{3}{2}$.
3. **General Solution:**
$$y = e^x(C_1\cos x+C_2\sin x)+e^{-x}(C_3\cos x+C_4\sin x)+\frac{x^4}{4}-\frac{3}{2}$$

24.3.4 Question 18

Solve: $(2D^2+3D+4)y = x^2-2x$

Solution:

1. **Find** y_c**:** $2m^2+3m+4=0 \implies m = \frac{-3\pm i\sqrt{23}}{4}$. $y_c = e^{-3x/4}\left(C_1\cos\frac{\sqrt{23}}{4}x+C_2\sin\frac{\sqrt{23}}{4}x\right)$.

2. **Find y_p:** $y_p = \frac{1}{4[1+\frac{3D+2D^2}{4}]}(x^2-2x) = \frac{1}{4}\left[1 - \frac{3D+2D^2}{4} + \frac{9D^2}{16}\right](x^2-2x)$. Simplify operator: $\frac{1}{4}\left[1 - \frac{3D}{4} + \frac{D^2}{16}\right](x^2-2x)$. Operate: $\frac{1}{4}\left[(x^2-2x) - \frac{3}{4}(2x-2) + \frac{1}{16}(2)\right] = \frac{1}{4}\left(x^2 - \frac{7x}{2} + \frac{13}{8}\right)$.
3. **General Solution:**
$$y = y_c + \frac{x^2}{4} - \frac{7x}{8} + \frac{13}{32}$$

24.3.5 Question 19

Solve: $(D^2 - 4D + 3)y = x^3$

Solution:

1. **Find y_c:** $m^2 - 4m + 3 = 0 \implies m = 1, 3$. $y_c = C_1e^x + C_2e^{3x}$.
2. **Find y_p:** $y_p = \frac{1}{3(1-\frac{4D-D^2}{3})}x^3 = \frac{1}{3}\left[1 + \left(\frac{4D-D^2}{3}\right) + \left(\frac{4D-D^2}{3}\right)^2 + \left(\frac{4D-D^2}{3}\right)^3\right]x^3$. Expand up to D^3: $\frac{1}{3}\left[1 + \frac{4D}{3} - \frac{D^2}{3} + \frac{16D^2}{9} - \frac{8D^3}{9} + \frac{64D^3}{27}\right] = \frac{1}{3}\left[1 + \frac{4D}{3} + \frac{13D^2}{9} + \frac{40D^3}{27}\right]$. Operate on x^3: $\frac{1}{3}\left[x^3 + \frac{4}{3}(3x^2) + \frac{13}{9}(6x) + \frac{40}{27}(6)\right] = \frac{1}{3}\left(x^3 + 4x^2 + \frac{26x}{3} + \frac{80}{9}\right)$.
3. **General Solution:**
$$y = C_1e^x + C_2e^{3x} + \frac{x^3}{3} + \frac{4x^2}{3} + \frac{26x}{9} + \frac{80}{27}$$

24.4 Part D: Mixed Cases & Superposition (Trig, Exponential, Algebraic)

24.4.1 Question 20

Solve: $(D^2 - 4D + 4)y = 8(e^{2x} + x^2 + \sin 2x)$

Solution:

1. **Find y_c:** $(m-2)^2 = 0 \implies m = 2, 2$. $y_c = (C_1 + C_2x)e^{2x}$.
2. **Find y_p (Superposition):** $y_p = y_{p1} + y_{p2} + y_{p3}$.
   - $y_{p1} = 8\frac{1}{(D-2)^2}e^{2x}$. Double failure at $D = 2$. $\implies 8\frac{x^2}{2}e^{2x} = 4x^2e^{2x}$.
   - $y_{p2} = 8\frac{1}{4(1-D+D^2/4)}x^2 = 2[1 + D + \frac{3D^2}{4}]x^2 = 2[x^2 + 2x + \frac{3}{4}(2)] = 2x^2 + 4x + 3$.
   - $y_{p3} = 8\frac{1}{D^2-4D+4}\sin 2x$. Substitute $D^2 = -4$: $\frac{8}{-4-4D+4}\sin 2x = \frac{8}{-4D}\sin 2x = -\frac{2}{D}\sin 2x = -2(-\frac{\cos 2x}{2}) = \cos 2x$.
3. **General Solution:**
$$y = (C_1 + C_2x)e^{2x} + 4x^2e^{2x} + 2x^2 + 4x + 3 + \cos 2x$$

24.4.2 Question 21

Solve: $\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 3y = e^{2x} + \cos x$

Solution:

1. **Find y_c:** $m^2 - 4m + 3 = 0 \implies m = 1, 3$. $y_c = C_1e^x + C_2e^{3x}$.
2. **Find y_p:**
   - $y_{p1} = \frac{1}{D^2-4D+3}e^{2x}$. Sub $D = 2$: $\frac{1}{4-8+3}e^{2x} = -e^{2x}$.

- $y_{p2} = \frac{1}{D^2-4D+3}\cos x$. Sub $D^2 = -1$: $\frac{1}{2-4D}\cos x$. Rationalize: $\frac{2+4D}{4-16D^2}\cos x$. Sub $D^2 = -1$: $\frac{2+4D}{20}\cos x = \frac{2\cos x - 4\sin x}{20} = \frac{\cos x - 2\sin x}{10}$.

3. **General Solution:**

$$y = C_1e^x + C_2e^{3x} - e^{2x} + \frac{1}{10}(\cos x - 2\sin x)$$

24.4.3 Question 22

Solve: $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + 3y = \cos x + x^2$

Solution:

1. **Find y_c:** $m^2 - 2m + 3 = 0 \implies m = \frac{2\pm\sqrt{4-12}}{2} = 1 \pm i\sqrt{2}$. $y_c = e^x(C_1\cos\sqrt{2}x + C_2\sin\sqrt{2}x)$.
2. **Find y_p:**
   - $y_{p1} = \frac{1}{D^2-2D+3}\cos x$. Sub $D^2 = -1$: $\frac{1}{2-2D}\cos x = \frac{2+2D}{4-4D^2}\cos x = \frac{2\cos x - 2\sin x}{8} = \frac{\cos x - \sin x}{4}$.
   - $y_{p2} = \frac{1}{3(1+\frac{D^2-2D}{3})}x^2 = \frac{1}{3}[1 - \frac{D^2-2D}{3} + \frac{4D^2}{9}]x^2 = \frac{1}{3}[1 + \frac{2D}{3} + \frac{D^2}{9}]x^2 = \frac{1}{3}(x^2 + \frac{4x}{3} + \frac{2}{9})$.
3. **General Solution:**

$$y = y_c + \frac{\cos x - \sin x}{4} + \frac{x^2}{3} + \frac{4x}{9} + \frac{2}{27}$$

24.5 Part E: Initial Value Problems

24.5.1 Question 23

Solve: $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} - 3y = 2e^{2x} + 10\sin 3x$, given that $y(0) = 2$ and $y'(0) = 4$.

Solution:

1. **Find y_c:** $m^2 - 2m - 3 = 0 \implies m = 3, -1$. $y_c = C_1e^{3x} + C_2e^{-x}$.
2. **Find y_p:**
   - $y_{p1} = \frac{2}{D^2-2D-3}e^{2x}$. Sub $D = 2$: $\frac{2}{4-4-3}e^{2x} = -\frac{2}{3}e^{2x}$.
   - $y_{p2} = \frac{10}{D^2-2D-3}\sin 3x$. Sub $D^2 = -9$: $\frac{10}{-12-2D}\sin 3x = \frac{-5}{D+6}\sin 3x = \frac{-5(D-6)}{D^2-36}\sin 3x = \frac{-5(D-6)}{-45}\sin 3x = \frac{1}{9}(3\cos 3x - 6\sin 3x) = \frac{1}{3}\cos 3x - \frac{2}{3}\sin 3x$.
3. **General Solution:**

$$y = C_1e^{3x} + C_2e^{-x} - \frac{2}{3}e^{2x} + \frac{1}{3}\cos 3x - \frac{2}{3}\sin 3x$$

4. **Apply Initial Conditions:** First, apply $y(0) = 2$: $2 = C_1(1) + C_2(1) - \frac{2}{3}(1) + \frac{1}{3}(1) - 0 \implies C_1 + C_2 = \frac{7}{3}$. *(Eq. A)* Next, find y': $y' = 3C_1e^{3x} - C_2e^{-x} - \frac{4}{3}e^{2x} - \sin 3x - 2\cos 3x$ Apply $y'(0) = 4$: $4 = 3C_1 - C_2 - \frac{4}{3} - 0 - 2 \implies 3C_1 - C_2 = \frac{22}{3}$. *(Eq. B)*
5. **Solve for Constants:** Add (Eq. A) and (Eq. B): $4C_1 = \frac{29}{3} \implies C_1 = \frac{29}{12}$. Substitute C_1 into (Eq. A): $C_2 = \frac{7}{3} - \frac{29}{12} = \frac{28}{12} - \frac{29}{12} = -\frac{1}{12}$.
6. **Final Particular Solution:**

$$y = \frac{29}{12}e^{3x} - \frac{1}{12}e^{-x} - \frac{2}{3}e^{2x} + \frac{1}{3}\cos 3x - \frac{2}{3}\sin 3x$$

Next Module: You have successfully learned how to solve linear equations with *constant* coefficients. But what happens if the coefficients are variables (like x^2 or x)? Proceed to Module 5: Cauchy-Euler Differential Equations.

25 Module 5: Higher-Order Linear ODEs (Variable Coefficients)

Welcome to **Module 5**. Up until this point, we have focused exclusively on higher-order linear differential equations where the coefficients were constants (e.g., $3y'' + 2y' + y = 0$).

However, in many advanced engineering and physics problems, the coefficients are variables—specifically, powers of x. In this module, we will learn how to handle a very specific and famous type of variable-coefficient equation: the **Cauchy-Euler Equation**.

25.1 Module Learning Objectives

By the end of this module, you will be able to:

1. Identify a Cauchy-Euler Differential Equation by checking if the power of the independent variable x matches the order of the derivative it multiplies.
2. Apply the independent variable substitution ($x = e^t$) to transform the variable-coefficient equation into a standard constant-coefficient equation.
3. Solve the transformed equation using the techniques mastered in Module 4, and translate the final answer back into terms of x.

25.2 Topic Index

25.2.1 1. Cauchy-Euler Differential Equations

- **The Concept:** Recognizing the $x^n \frac{d^n y}{dx^n}$ structure.
- **The Substitution Rule:** Letting $x = e^t$ and defining the new operator $D = \frac{d}{dt}$.
- **The Transformation:** Changing $x\frac{dy}{dx}$ to Dy and $x^2\frac{d^2y}{dx^2}$ to $D(D-1)y$.
- *Includes step-by-step solved questions converting back and forth between x and t domains.*

25.2.2 2. Practice Questions

- Comprehensive Cauchy-Euler problems including algebraic, logarithmic, and trigonometric right-hand sides.
- *Includes fully worked solutions for 4th order, complex roots, and double failure cases.*

26 1. Cauchy-Euler Differential Equations

In the previous module, we solved Higher-Order Linear ODEs where the coefficients were constants. Now, we will look at a special type of linear equation where the coefficients are variables—specifically, powers of x.

This specific form is known as the **Cauchy-Euler Equation** (or Cauchy's Differential Equation).

26.1 1. Concept Overview

A Cauchy-Euler equation has a very recognizable signature: **the power of the independent variable x exactly matches the order of the derivative it is multiplied by.**

26.1.1 The Standard Form

$$a_n x^n \frac{d^n y}{dx^n} + a_{n-1} x^{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1 x \frac{dy}{dx} + a_0 y = F(x)$$

Where $a_0, a_1, \ldots, a_n$ are constants.

Notice that x^2 is paired with $\frac{d^2y}{dx^2}$, x^1 is paired with $\frac{dy}{dx}$, and so on.

26.2 2. Working Rule (The $x = e^t$ Substitution)

We cannot solve this directly using our constant-coefficient methods. However, we can use a clever substitution to transform the independent variable from x to t, which magically turns the variable coefficients into constant coefficients!

Step 1: The Core Substitution Let $x = e^t$. This also means that $t = \ln x$.

Step 2: Define the New Operator Let $D = \frac{d}{dt}$ (Notice the derivative is now with respect to t, not x).

Step 3: Transform the Derivatives Using the chain rule (proof omitted for brevity), the x-derivatives transform into D-operators as follows:

- $x\frac{dy}{dx} = Dy$
- $x^2\frac{d^2y}{dx^2} = D(D-1)y = (D^2 - D)y$
- $x^3\frac{d^3y}{dx^3} = D(D-1)(D-2)y = (D^3 - 3D^2 + 2D)y$

Step 4: Substitute and Solve Substitute these operators into the original equation. It will now be a standard Higher-Order Linear ODE with constant coefficients. Solve for $y(t) = y_c + y_p$ using the methods from Module 4.

Step 5: Resubstitute back to x Once you have the final answer in terms of t, replace every t with $\ln x$ and every e^{kt} with x^k.

26.3 3. Solved Questions

Here are two comprehensive examples from the lecture notes demonstrating this transformation.

26.3.1 Question 1: Algebraic Right-Hand Side

Solve: $x^2\frac{d^2y}{dx^2} + 7x\frac{dy}{dx} + 5y = x^5$

Solution:

Step 1: Apply Transformations Let $x = e^t$, $t = \ln x$, and $D = d/dt$. Substitute the operators into the left side, and e^t into the right side:

$$[D(D-1) + 7D + 5]y = (e^t)^5$$
$$(D^2 - D + 7D + 5)y = e^{5t}$$
$$(D^2 + 6D + 5)y = e^{5t}$$

(This is now a constant-coefficient equation!)

Step 2: Find the Complementary Function (y_c) Auxiliary Equation: $m^2+6m+5 = 0$ Factor: $(m+5)(m+1) = 0 \implies m = -1, -5$

$$y_c = C_1e^{-t} + C_2e^{-5t}$$

Step 3: Find the Particular Integral (y_p)

$$y_p = \frac{1}{D^2 + 6D + 5}e^{5t}$$

Apply the Exponential Rule ($D \to 5$):

$$y_p = \frac{1}{5^2 + 6(5) + 5}e^{5t} = \frac{1}{25 + 30 + 5}e^{5t} = \frac{1}{60}e^{5t}$$

Step 4: Write General Solution in t

$$y(t) = y_c + y_p = C_1e^{-t} + C_2e^{-5t} + \frac{1}{60}e^{5t}$$

Step 5: Resubstitute back to x Recall that $e^t = x$, so $e^{-t} = x^{-1}$ and $e^{5t} = x^5$.

$$y(x) = C_1x^{-1} + C_2x^{-5} + \frac{x^5}{60}$$

26.3.2 Question 2: Trigonometric & Shift Rule Combination

Solve: $x^2\frac{d^2y}{dx^2} - 3x\frac{dy}{dx} + 5y = x^2\sin(\ln x)$

Solution:

Step 1: Apply Transformations Let $x = e^t$, $t = \ln x$, and $D = d/dt$. Substitute into the equation:

$$[D(D-1) - 3D + 5]y = (e^t)^2\sin(t)$$
$$(D^2 - D - 3D + 5)y = e^{2t}\sin t$$
$$(D^2 - 4D + 5)y = e^{2t}\sin t$$

Step 2: Find the Complementary Function (y_c) Auxiliary Equation: $m^2-4m+5 = 0$ Use quadratic formula: $m = \frac{4\pm\sqrt{16-20}}{2} = \frac{4\pm 2i}{2} = 2 \pm i$ Complex roots ($\alpha = 2, \beta = 1$):

$$y_c = e^{2t}(C_1\cos t + C_2\sin t)$$

Step 3: Find the Particular Integral (y_p)

$$y_p = \frac{1}{D^2 - 4D + 5}(e^{2t} \sin t)$$

Apply Exponential Shift Rule (Shift e^{2t} to the front, replace $D \to D + 2$):

$$y_p = e^{2t} \frac{1}{(D + 2)^2 - 4(D + 2) + 5} \sin t$$

$$y_p = e^{2t} \frac{1}{(D^2 + 4D + 4) - 4D - 8 + 5} \sin t$$

$$y_p = e^{2t} \frac{1}{D^2 + 1} \sin t$$

Step 4: Solve Trigonometric part (Case of Failure) Apply Trig Rule for $\sin(1t)$ (Replace $D^2 \to -1^2 = -1$): Denominator becomes $-1 + 1 = 0$. This is a Case of Failure. Multiply by t (our new independent variable) and differentiate denominator ($2D$):

$$y_p = e^{2t} \cdot t \cdot \frac{1}{2D} \sin t$$

The operator $\frac{1}{D}$ means integrate with respect to t:

$$y_p = \frac{te^{2t}}{2} \int \sin t \, dt = \frac{te^{2t}}{2}(-\cos t)$$

$$y_p = -\frac{t}{2} e^{2t} \cos t$$

Step 5: Write General Solution in t

$$y(t) = e^{2t}(C_1 \cos t + C_2 \sin t) - \frac{t}{2} e^{2t} \cos t$$

Step 6: Resubstitute back to x Substitute $t = \ln x$ and $e^{2t} = x^2$:

$$y(x) = x^2[C_1 \cos(\ln x) + C_2 \sin(\ln x)] - \frac{1}{2}(\ln x)x^2 \cos(\ln x)$$

Next Steps: Practice these transformations with the Module 5 Practice Questions.

27 Module 5 Practice: Cauchy-Euler Equations

This section contains all the comprehensive homework and practice problems for variable-coefficient linear differential equations (Cauchy-Euler equations).

Core Substitution Rule Reminder: Let $x = e^t$, which means $t = \ln x$. Let $D = \frac{d}{dt}$. Transformations:

- $x \frac{dy}{dx} = Dy$
- $x^2 \frac{d^2y}{dx^2} = D(D - 1)y = (D^2 - D)y$
- $x^3 \frac{d^3y}{dx^3} = D(D - 1)(D - 2)y = (D^3 - 3D^2 + 2D)y$
- $x^4 \frac{d^4y}{dx^4} = D(D - 1)(D - 2)(D - 3)y = (D^4 - 6D^3 + 11D^2 - 6D)y$

27.0.1 Question 1 (From Whiteboard Q1)

Solve: $x^2\frac{d^2y}{dx^2} - 2x\frac{dy}{dx} - 4y = x^2$

Solution:

1. **Apply Transformations:** Substitute $x^2\frac{d^2y}{dx^2} = (D^2 - D)y$, $x\frac{dy}{dx} = Dy$, and $x^2 = (e^t)^2 = e^{2t}$.
$$(D^2 - D)y - 2Dy - 4y = e^{2t}$$
$$(D^2 - 3D - 4)y = e^{2t}$$
2. **Find Complementary Function (y_c):** Auxiliary Equation: $m^2 - 3m - 4 = 0$ Factorizing: $(m-4)(m+1) = 0 \implies m = 4, -1$
$$y_c(t) = C_1e^{4t} + C_2e^{-t}$$
3. **Find Particular Integral (y_p):**
$$y_p(t) = \frac{1}{D^2 - 3D - 4}e^{2t}$$
Substitute $D = 2$:
$$y_p(t) = \frac{1}{(2)^2 - 3(2) - 4}e^{2t} = \frac{1}{4 - 6 - 4}e^{2t} = -\frac{1}{6}e^{2t}$$
4. **General Solution in terms of t:**
$$y(t) = C_1e^{4t} + C_2e^{-t} - \frac{1}{6}e^{2t}$$
5. **Resubstitute back to x ($e^t = x$):**
$$y(x) = C_1x^4 + C_2x^{-1} - \frac{1}{6}x^2$$

27.0.2 Question 2 (From Whiteboard Q2)

Solve: $x^2\frac{d^2y}{dx^2} - 2x\frac{dy}{dx} + 2y = x^3$

Solution:

1. **Apply Transformations:**
$$(D^2 - D)y - 2Dy + 2y = (e^t)^3$$
$$(D^2 - 3D + 2)y = e^{3t}$$
2. **Find Complementary Function (y_c):** Auxiliary Equation: $m^2 - 3m + 2 = 0$ Factorizing: $(m-1)(m-2) = 0 \implies m = 1, 2$
$$y_c(t) = C_1e^t + C_2e^{2t}$$

3. **Find Particular Integral (y_p):**

$$y_p(t) = \frac{1}{D^2 - 3D + 2}e^{3t}$$

Substitute $D = 3$:

$$y_p(t) = \frac{1}{(3)^2 - 3(3) + 2}e^{3t} = \frac{1}{9 - 9 + 2}e^{3t} = \frac{1}{2}e^{3t}$$

4. **General Solution in terms of t:**

$$y(t) = C_1e^t + C_2e^{2t} + \frac{1}{2}e^{3t}$$

5. **Resubstitute back to x ($e^t = x$):**

$$y(x) = C_1x + C_2x^2 + \frac{1}{2}x^3$$

27.0.3 Question 3 (From Whiteboard Q3 - 4th Order)

Solve: $x^4\frac{d^4y}{dx^4} + 6x^3\frac{d^3y}{dx^3} + 9x^2\frac{d^2y}{dx^2} + 3x\frac{dy}{dx} + y = (1 + \ln x)^2$

Solution:

1. **Apply Transformations:** Let $x = e^t \implies t = \ln x$. The right side becomes $(1 + t)^2 = t^2 + 2t + 1$. Substitute the operator equivalents into the left side: $[(D^4 - 6D^3 + 11D^2 - 6D) + 6(D^3 - 3D^2 + 2D) + 9(D^2 - D) + 3D + 1]y = t^2 + 2t + 1$ Expand the brackets: $(D^4 - 6D^3 + 11D^2 - 6D + 6D^3 - 18D^2 + 12D + 9D^2 - 9D + 3D + 1)y$ Group like terms: $D^4 + (-6 + 6)D^3 + (11 - 18 + 9)D^2 + (-6 + 12 - 9 + 3)D + 1$ $(D^4 + 2D^2 + 1)y = t^2 + 2t + 1$ Notice that $(D^4 + 2D^2 + 1)$ is a perfect square: $(D^2 + 1)^2$.

$$(D^2 + 1)^2y = t^2 + 2t + 1$$

2. **Find Complementary Function (y_c):** Auxiliary Eq: $(m^2 + 1)^2 = 0 \implies m^2 = -1 \implies m = \pm i$ (Repeated twice)

$$y_c(t) = (C_1 + C_2t)\cos t + (C_3 + C_4t)\sin t$$

3. **Find Particular Integral (y_p):** Because the RHS is algebraic, we use binomial expansion.

$$y_p(t) = \frac{1}{(1 + D^2)^2}(t^2 + 2t + 1) = (1 + D^2)^{-2}(t^2 + 2t + 1)$$

Expand $(1 + x)^{-2} = 1 - 2x + 3x^2 - \ldots$

$$y_p(t) = (1 - 2D^2 + 3D^4 - \ldots)(t^2 + 2t + 1)$$

Apply the operator (derivatives higher than D^2 on a t^2 polynomial will be 0):

$$y_p(t) = 1(t^2 + 2t + 1) - 2D^2(t^2 + 2t + 1)$$

First derivative: $2t + 2$. Second derivative (D^2): 2.

$$y_p(t) = (t^2 + 2t + 1) - 2(2) = t^2 + 2t + 1 - 4 = t^2 + 2t - 3$$

4. **General Solution in terms of t:**

$$y(t) = (C_1 + C_2t)\cos t + (C_3 + C_4t)\sin t + t^2 + 2t - 3$$

5. **Resubstitute back to x ($t = \ln x$):**

$$y(x) = (C_1 + C_2 \ln x)\cos(\ln x) + (C_3 + C_4 \ln x)\sin(\ln x) + (\ln x)^2 + 2\ln x - 3$$

27.0.4 Question 4 (From Notes - Logarithmic RHS)

Solve: $x^2 \frac{d^2y}{dx^2} - x\frac{dy}{dx} + y = \ln x$

Solution:

1. **Apply Transformations:**

$$[D(D-1) - D + 1]y = t$$

$$(D^2 - 2D + 1)y = t \implies (D-1)^2 y = t$$

2. **Find Complementary Function (y_c):** Auxiliary Eq: $(m-1)^2 = 0 \implies m = 1, 1$

$$y_c(t) = (C_1 + C_2t)e^t$$

3. **Find Particular Integral (y_p):**

$$y_p(t) = \frac{1}{D^2 - 2D + 1}t = \frac{1}{1 - (2D - D^2)}t$$

Binomial Expansion: $[1 - (2D - D^2)]^{-1}t \approx (1 + 2D)t$

$$y_p(t) = t + 2D(t) = t + 2(1) = t + 2$$

4. **General Solution in terms of t:**

$$y(t) = (C_1 + C_2t)e^t + t + 2$$

5. **Resubstitute back to x ($e^t = x, t = \ln x$):**

$$y(x) = x(C_1 + C_2 \ln x) + \ln x + 2$$

27.0.5 Question 5 (From Notes - Complex Roots)

Solve: $x^2y'' + y = 3x^2$

Solution:

1. **Apply Transformations:**

$$(D^2 - D)y + y = 3e^{2t}$$

$$(D^2 - D + 1)y = 3e^{2t}$$

2. **Find Complementary Function (y_c):** Auxiliary Eq: $m^2 - m + 1 = 0$ Quadratic Formula: $m = \frac{1 \pm \sqrt{1-4}}{2} = \frac{1 \pm i\sqrt{3}}{2} = \frac{1}{2} \pm i\frac{\sqrt{3}}{2}$
$$y_c(t) = e^{t/2}\left(C_1 \cos\left(\frac{\sqrt{3}}{2}t\right) + C_2 \sin\left(\frac{\sqrt{3}}{2}t\right)\right)$$
3. **Find Particular Integral (y_p):**
$$y_p(t) = \frac{1}{D^2 - D + 1} 3e^{2t}$$
Substitute $D = 2$:
$$y_p(t) = 3\frac{1}{2^2 - 2 + 1}e^{2t} = 3\frac{1}{4 - 2 + 1}e^{2t} = 3\left(\frac{1}{3}\right)e^{2t} = e^{2t}$$
4. **General Solution in terms of t:**
$$y(t) = e^{t/2}\left(C_1 \cos\left(\frac{\sqrt{3}}{2}t\right) + C_2 \sin\left(\frac{\sqrt{3}}{2}t\right)\right) + e^{2t}$$
5. **Resubstitute back to x ($e^t = x, e^{t/2} = x^{1/2} = \sqrt{x}$):**
$$y(x) = \sqrt{x}\left(C_1 \cos\left(\frac{\sqrt{3}}{2}\ln x\right) + C_2 \sin\left(\frac{\sqrt{3}}{2}\ln x\right)\right) + x^2$$

27.0.6 Question 6 (From Notes - Double Failure Case)

Solve: $x^2y'' - 3xy' + 4y = 2x^2$

Solution:

1. **Apply Transformations:**
$$[D(D-1) - 3D + 4]y = 2e^{2t}$$
$$(D^2 - 4D + 4)y = 2e^{2t} \implies (D-2)^2 y = 2e^{2t}$$
2. **Find Complementary Function (y_c):** Auxiliary Eq: $(m-2)^2 = 0 \implies m = 2, 2$
$$y_c(t) = (C_1 + C_2 t)e^{2t}$$
3. **Find Particular Integral (y_p):**
$$y_p(t) = \frac{1}{(D-2)^2} 2e^{2t}$$
Substitute $D = 2 \implies$ Denominator is 0 (Failure). *Fix 1:* Multiply by t, diff denominator: $y_p(t) = t\frac{1}{2(D-2)}2e^{2t}$. Sub $D = 2 \implies$ Failure again. *Fix 2:* Multiply by t again (t^2), diff denominator again: $y_p(t) = t^2\frac{1}{2(1)}2e^{2t} = t^2e^{2t}$.
4. **General Solution in terms of t:**
$$y(t) = (C_1 + C_2 t)e^{2t} + t^2e^{2t}$$
5. **Resubstitute back to x:**
$$y(x) = (C_1 + C_2 \ln x)x^2 + (\ln x)^2 x^2$$

Congratulations! You have completed the structured course notes for Ordinary Differential Equations based on your provided material.